Physics from Existence I: The Equation

The Binary-Entropy Equation $V = -H$ and the Structure of Standard-Model Parameters

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Abstract

No theory has simultaneously derived all 26 free parameters of the Standard Model. In this paper, extending the well-known quantum indefiniteness of existence to the existence of the universe itself, we posit three axioms about existence—existence is bivalent ($n \in \{0,1\}$, i.e. $n^2{=}n$), existence is uncertain ($\sigma{\neq}1$), and non-existence is forbidden ($\sigma{\neq}0$)—and show that they uniquely determine the single equation $V = -H$ as an algebraic identity, and that the mathematical structure derived from its exact algebraic expansion corresponds to the structures of the Standard Model. As a result, 26 physical constants in total are reproduced with no adjustable parameters. For those with established experimental values, the agreement is 0.0006%–2.6% for the twenty derived in this paper.

Derived from the axioms of existence, $V = -H$ is the canonical free energy of a single binary degree of freedom $n \in \{0,1\}$, determined as the Legendre transform of the Fermi–Dirac distribution. From this potential and the kinetic term of the canonical normalisation ($g = 1$, §3.1), the Schrödinger equation is derived as a transfer matrix, and we show that it has exactly three bound states (discrete solutions, $N = 3$), corresponding to the three generations of fermions. The spatial dimension $d = 3$ follows from three conditions—preservation of the well, spin-statistics, and propagating gravity—and the Lie-algebra structure unfolding from this double coincidence ($N = d = 3$) reproduces the gauge group SU(3)×SU(2)×U(1).

Twenty-six physical constants are derived from this structure, including $1/\alpha = 137.035999084$ (all orders, matching CODATA 2018 to all displayed digits; the leading term is 137.0368 at 0.0006%), $\alpha_s = 0.1179$ (0.07%), $\sin^2\theta_W = 0.23122$ ($< 0.01\%$), the charged-lepton mass ratio $R = 1.5293$ (0.006%), and $\sin\theta_C = 0.2245$ (0.10%). The results further predict the absolute neutrino masses ($\Sigma m_\nu = 59.50$ meV once the absolute scale is set by Δm^2_{21}) and strictly zero neutrinoless double-beta decay, among other testable predictions. These quantities are not independent input parameters—they are all computed from the same mathematical structure fixed by $V = -H$.

All of these numerical values are reproducible with the attached verification code.

1 Introduction

The Standard Model is the most successful theory of particle physics. It describes all known particle interactions and has withstood extensive experimental tests, including the discovery of the Higgs boson [1, 2] (2012). At present, however, the theory contains 26 free parameters[1]—fermion masses (Yukawa couplings), mixing angles, gauge couplings, the Higgs self-coupling, and CP phases. Their values can only be put in from measurement; why these physical constants take the values they do is not explained within the Standard Model. The generation number N = 3 is likewise an unexplained input.

Despite many attempts—grand unified theories (GUTs), supersymmetry (SUSY), string theory—no theory has simultaneously derived all 26 parameters with zero free parameters [3]. Indeed, even for the single constant α, the fine-structure constant, attempts at a first-principles derivation [35, 36, 37] have a long history, yet no theory has derived it from an equation with a unifying mechanism.

Approaches that derive physics from information theory have also been attempted. Wheeler's "it from bit" [4] proposed the vision that physical reality arises from information, but had no mathematical framework. Jaynes [5] derived statistical mechanics from entropy maximisation, but did not reach quantum field theory. Verlinde [7] derived gravity from entropic reasoning, but gravity alone. There are also Frieden's Fisher-information approach [6], Caticha's information geometry [8], and the information-theoretic axiomatisations of Hardy [9] and Chiribella et al. [10]. The prior work closest in spirit is the Connes–Chamseddine spectral Standard Model [39, 40, 41], which derives the structure of the gauge group and the Higgs sector from noncommutative geometry (a related algebraic reconstruction is Furey's division-algebra approach [42]). All these approaches share a common limitation: they can derive the *form* of physical equations, but not the *values* of the constants appearing in them.

Now, "Why do I exist? Why does the universe exist?"—who has not, at some point, let their thoughts dwell on such questions? Traditionally, they have been regarded as purely philosophical questions, beyond the scope of physics. In fact, however, quantum theory has already established a decisive fact about the very notion of existence. As is well known, the existence of a quantum object—whether a mode is occupied—is in general not definite, and is given only as a probability. In other words, at the microscopic level, existence is not a settled given. If so, might the "existence" of the universe itself perhaps behave in the same way? What happens if that question is written down as an equation? That is the starting point of this paper.

This paper first formalises this question about the existence of the universe as three axioms—Existence (n $\in$ 0,1), Uncertainty ($\sigma \neq 1$), and Non-existence is forbidden ($\sigma \neq 0$). Together A2 and A3 determine $\sigma \in (0,1)$, a partition function $\mathrm{Z} = 1 + e^{\varphi}$, and, via the Legendre transform of the Fermi–Dirac distribution, the canonical free energy $\mathrm{V} = -\mathrm{H}(\sigma(\phi))$ as an algebraic identity, where H is the binary Shannon entropy, $\sigma(\phi)$ is the Fermi–Dirac distribution (sigmoid function), and $\phi = \mathrm{logit}(\sigma)$ is the natural parameter [14, 15].

Combining this potential $\mathrm{V} = -\mathrm{H}(\sigma(\phi))$ with the kinetic term of the canonical normalisation (g = 1, §3.1), the Schrödinger equation is derived as a transfer matrix. Carrying the computation further, this Schrödinger equation has exactly three bound states (N =

[1] 19 in the minimal Standard Model with massless neutrinos; 26 with three neutrino masses and four PMNS parameters. Some references count 25 by absorbing v_{EW} into the energy scale.

3), and a structure corresponding to the three generations of fermions appears. At the same time, the spatial dimension d = 3 follows from three conditions—preservation of the well, spin-statistics, and propagating gravity. The spacetime dimension is D = d + 1 = 4, supported as consistency checks under stated premises by the gravitational degree-of-freedom count, by $\mathbb{C}P^2$, and by the connection of the signature (3,1) to the $M_2(\mathbb{C})_{sa}$ structure (§3.4, §8). From the coincidence of these two numbers (N = d = 3) the finite geometry PG(2, F_3) is fixed, and the Lie-algebra structure unfolding from its 9+3+1 partition corresponds uniquely to the structure of the gauge group SU(3)×SU(2)×U(1).

From the representations of this gauge group, a structure yielding all the particles—48 fermions, 12 gauge bosons, and 1 Higgs—is derived (§4). From this structure, 20 parameters are derived—including $1/\alpha_{em}$ = 137.0368 (0.0006%; 137.035999084 at all orders—proved in Paper II), $\sin^2\theta_W = 3/13$ (0.2%), the charged-lepton mass ratio R = 1.5293 (0.006%), Koide's Q ≈ 3/2 ($\delta = -1.5 \times 10^{-5}$), and CKM/PMNS mixing angles—all at 0.0006%–2.6% accuracy with zero adjustable parameters. The six derived in Paper II ($|V_{cb}|$, $|V_{ub}|$, δ_{CKM}, δ_{PMNS}, m_3/m_2, $\Delta m^2{}_{31}/\Delta m^2{}_{21}$) are also displayed in the tables (§9). The associated Paper II reconstruction additionally gives the matching-point condition $\bar{\theta}(\mu_0) = 0$[II].

Thus the algebraic expansion of "existence" alone yields numbers that agree with the Standard-Model constants to high accuracy; beyond this numerical agreement, the mathematical structure derived from V = −H also corresponds to the structural problems of the Standard Model themselves. For example, the role played in the Standard Model by $\mu^2 > 0$—moving the vacuum from the symmetric point to a finite field value—is played here by the shape of the well of V (§6); the bounded internal potential (V ∈ [−ln 2, 0]) supplies no instability channel, so neither the metastability problem nor the hierarchy problem arises within this construction (the correspondence to 4D vacuum stability and the hierarchy problem requires the reconstruction matching). Since V = −H is uniquely determined as the canonical free energy of n ∈ 0,1, the Higgs is fixed to be unique.

What this paper shows is that, when the mathematical structure generated by V = −H is placed in correspondence with the structure of the Standard Model, the derived numbers agree with experiment—this paper itself does not carry out a rigorous first-principles derivation of the Standard Model; that derivation is treated in Paper II as a theorem under the stated realisation conditions (§8.2). The method of computing the physical constants from the derived equations alone is given in this paper, and can be reproduced independently with the attached verification code.

Note that comparing the derived numbers with the experimental observables requires making two things explicit. First, dimensionful quantities require a correspondence to humanly chosen units of measurement—setting the absolute scale v (§9). Second, dimensionless quantities require a correspondence that identifies which part of the structure is which physical structure or observable—for example, reading and identifying $\sin^2\theta_W$ as the fraction of the 13 directions occupied by the 3 weak ones. In other words, one must identify which known physical structure each derived mathematical structure corresponds to.

Each of these identifications is provisional at the stage of its declaration. The text declares each identification where it enters (§3 generations, §4.2 gauge sectors, §6 Higgs, §7 masses and mixings) and argues under that hypothesis; the agreement of the full set of derived values with experiment is what corroborates the identifications. This paper states all of these correspondences (the mapping rules onto the observables) explicitly and computes the values. That each mapping rule is not a free parameter—the proof of its uniqueness—together with the operator-level derivations of the individual constructions, would require an enormous body of argument and is therefore deferred to Paper II [20]. Such claims are marked with a superscript $^{\mathrm{II}}$ in the text (indicating rigorisation in Paper II) and are not restated individually.

The paper is organized as follows. Section 2 derives $V = -H$ from three axioms. Section 3 establishes the Schrödinger equation, proves the bound-state count $N = 3$ as a mathematical theorem, and derives the spatial dimension $d = 3$ as the unique common solution of three independent realisation-consistency conditions. Section 4 constructs the gauge structure from $\mathrm{PG}(2, \mathbb{F}_3)$. Section 5 derives the gauge coupling constants. Section 6 derives the Higgs mass and shows that the vacuum is pinned at an interior point with the internal potential bounded (no instability channel). Section 7 identifies the bound states with the three generations of charged leptons and derives fermion masses and mixing. Section 8 confirms the spacetime dimension $D = 4$ and signature $(3, 1)$ by consistency checks with stated premises, and establishes the tensor-product structure $\mathcal{H}_{\mathrm{int}}(\phi) \otimes \mathcal{H}_{\mathrm{space}}(x)$ from the independence of the construction. Section 9 summarises all results and presents falsifiable predictions. Section 10 gives the conclusion.

2 From Existence to the Equation

This chapter formalises three axioms about existence mathematically and shows that the fundamental equation $V = -H$ is derived as an algebraic identity. From this equation, the geometric representation corresponding to the internal-symmetry structure of the Standard Model is uniquely determined$^{\mathrm{II}}$.

2.1 The Axioms

Paring the question of the introduction—might the existence of the universe itself, like quantum existence, be undetermined?—down to a form that can become an equation leaves three short sentences. What the three sentences speak of is a single proposition, prior to any property: that something exists. Which particles, what space, what laws—nothing of that is said yet. These three sentences are the entire input of the theory; no physical constant, no spacetime, and no postulate of quantum mechanics enters here.

A1 (Existence): Something exists. “Exists” is a proposition; it is either true or false — by the law of excluded middle, there is no third value. Writing the state of existence as n, the state space is $n \in \{0, 1\}$. A1 defines the state space but does not determine the **value** of n.

What is bivalent is the *proposition* of existence, not a claim that the world is deterministic. To ask about the existence of something is, before asking about any of its properties, the single question: is it, or is it not? A1 fixes only the minimal structure this question carries—a one-bit state space. Everything that follows is built on this one bit.

A2 (Uncertainty): Existence is uncertain. Whatever exists does not exist with certainty: the occupation probability $\sigma = \langle n \rangle$ satisfies $\sigma \neq 1$. A2 excludes the upper endpoint (certain existence) but says nothing about the lower endpoint.

If existence were certain ($\sigma = 1$), it would be an immovable background: nothing left to ask, nothing left to happen. As quantum theory teaches, microscopic existence is given only as a probability—A2 lifts this fact to the existence of the universe itself. Existence is not a given but something that *occurs*, carrying uncertainty. This one sentence is what later generates the statistical ensemble and the dynamics.

A3 (Prohibition of non-existence): Non-existence is not a state ($\sigma \neq 0$). A way of being that consists in not-being is a contradiction.

"Nothing" is not a way for something to be. If $\sigma = 0$ were a realisation state, non-existence would be *there*, carrying data—the absence of being would be. A3 is the hardest-working of the three axioms: here it only removes the lower endpoint, but later it selects the coordinate by forbidding model data at the point of non-existence (§3.1) and selects the spatial dimension as the prohibition of boundary conditions (§3.4).

The three axioms are mutually independent—A1 removes the third truth value, A2 the upper endpoint, A3 the lower, each removing what the other two do not. And the three are exhaustive: together they leave only the open interval $0 < \sigma < 1$ (§2.2), and no further assumption enters anywhere in this paper. The axioms are these three sentences; how they are realised mathematically is constructed in minimal form in the following sections, and the proof that no freedom of choice remains in that realisation is given in Paper II[II].

2.2 Bivalence and the Partition Function

Starting from the state space $n \in \{0, 1\}$ established by A1, we construct the partition function.

Since n takes only the values 0 and 1,

$$n^2 = n \tag{1}$$

follows — because the solutions of $n(n-1) = 0$ are exactly $\{0, 1\}$. However, A1 alone leaves the system in a pure state $n = 0$ or $n = 1$, with no dynamics. A2 ($\sigma \neq 1$) excludes certain existence; A3 ($\sigma \neq 0$) excludes non-existence. Together they determine $\sigma \in (0, 1)$: the system is described as a canonical ensemble of a binary variable—an exponential family with mean parameter σ—neither certainly present nor certainly absent. The tilt weight $e^{n\phi}$ has its tilt coefficient fixed to 1 (the scale of ϕ; §3.1)[II]. $n^2 = n$ truncates the sum to two terms, and the partition function is

$$Z = \sum_{n=0}^{1} e^{n\phi} = 1 + e^{\phi}, \tag{2}$$

where $\phi = \ln[\sigma/(1-\sigma)]$ (the logit transform) is the canonical coordinate of the exponential family.

Were n able to take values $0, 1, 2, \ldots$ (Bose statistics), the sum would be an infinite series. As a consequence of the law of excluded middle, $n^2 = n$ forbids this and restricts the occupation number to $\{0, 1\}$—the Fermi–Dirac occupation statistics. Exchange anti-symmetry (spin statistics) does not follow from this alone; it is carried by the many-body realisation (§4.2).

The expectation $\sigma = \langle n \rangle = \partial \ln Z / \partial \phi$ coincides with the Fermi–Dirac distribution, and its variance is the source of all subsequent structure:

$$\sigma = \frac{1}{1 + e^{-\phi}}, \qquad g_F = \sigma(1-\sigma) = \mathrm{Var}(n) > 0. \tag{3}$$

These are not assumed but derived, from the combination of A1 (bivalence), A2 (uncertainty: $\sigma \neq 1$), and A3 (non-existence forbidden: $\sigma \neq 0$). g_F is the gap between the binary observable ($n^2 = n$) and its uncertain expectation ($\sigma^2 \neq \sigma$), and vanishes only at the pure values $\sigma \in \{0, 1\}$.

2.3 The Effective Potential $V = -H$

§2.2 established the partition function $Z = 1 + e^{\phi}$. The next step is to construct an effective potential from it.

The naive choice $-\ln Z$ diverges as $\phi \to \infty$, so a Legendre transform to the canonical free energy is required. The logarithm $W = \ln Z$ of the partition function generates the classical field $\sigma = dW/d\phi$ and yields the effective potential:

$$V(\phi) = \phi\sigma - W(\phi) = \sigma \ln \sigma + (1-\sigma) \ln(1-\sigma) = -H(\sigma), \tag{4}$$

where $H(p) = -p \ln p - (1-p) \ln(1-p)$ is binary Shannon entropy [12].

That is, written out with its arguments, $V(\phi) = -H(\sigma(\phi))$: at each point of the canonical coordinate ϕ, the value of the potential equals minus the binary entropy of the occupation probability $\sigma(\phi)$. This is an algebraic identity—the canonical free energy of a single binary degree of freedom $n \in \{0, 1\}$ equals the negative of the Shannon entropy—and there is no freedom of choice. The bare form $V = -H$, used throughout, abbreviates this identity of functions. The potential

$$V = -H \tag{5}$$

is thereby uniquely determined by axioms A1–A3 together with the exponential-family realisation above.

From this single equation, a structure closely matching the Standard Model emerges, and values agreeing with physical constants to high precision are derived (§3–§9).

The derivation chain of this chapter is summarised as:

$$\text{A1–A3} \;\Rightarrow\; n^2{=}n,\; 0 < \sigma < 1 \;\Rightarrow\; Z = 1{+}e^{\phi} \;\xrightarrow{\text{Legendre}}\; V = -H$$

3 Derivation of the Schrödinger Equation and $N = d = 3$

$V = -H$ is a potential; by itself it does not yet say how many discrete states exist. This chapter first derives the eigenvalue equation—the Schrödinger equation—from the same exponential-family structure as the potential (§3.1). The spectral question—how many bound states the well holds—then becomes well-posed, and its answer is $N = 3$ (§3.2–§3.3). At the end of §3.3 the three bound states are identified with the three generations of fermions—the one physical identification this chapter makes. Only under that identification does the second question arise: the particles the bound states represent exist in real space, so what determines the dimension of that space? Realisation consistency answers it, and $d = 3$ follows (§3.4). The derivation of $d = 3$ makes no use of the value $N = 3$—the two derivations share the operator and nothing else. The coincidence $N = d = 3$ becomes the input of §4.

3.1 Derivation of the Schrödinger Equation

To write the eigenvalue equation, a kinetic term is needed in addition to the potential of §2.3, and it follows from the same structure as the potential. Two data remain: the coordinate carrying the motion, and one dimensionless normalisation. Both are fixed uniquely; the uniqueness proofs are deferred to Paper II[II].

First, the coordinate. The exponential family of a binary variable has two natural coordinates: $u = 2\arcsin\sqrt{\sigma}$, in which the Fisher metric is flat, and $\phi = \mathrm{logit}(\sigma)$, the canonical parameter in the exponent [13, 14, 15]. A3 decides between them. The coordinate u places the forbidden point $\sigma = 0$ at the finite endpoint $u = 0$, so dynamics on u would need boundary data exactly at the point of non-existence—non-existence would become a repository of model-defining data. The coordinate ϕ sends $\sigma = 0$ to $\phi \to -\infty$: no boundary data attaches to non-existence, and under A3 none is admissible. Among the admissible coordinates, ϕ is the one on which the tilting action of A1 is linear—unique up to affine maps—and the binary indicator $n \in \{0, 1\}$ fixes its scale. Free motion on ϕ therefore has a constant-coefficient quadratic Lagrangian; position-dependent coefficients are excluded by displacement homogeneity[II].

Second, the normalisation. Writing the kinetic coefficient as $a^2/2$ and the action weight as b, the apparent two-parameter family reduces to the single dimensionless ratio $g = b/a^2$: the binary indicator $n \in \{0, 1\}$ of A1 fixes the unit of ϕ, the Legendre identity fixes the unit of $V = -H$, and rescaling the transfer step moves a^2 and b in the same proportion. Measuring the quadratic variation per step and the action weight as quantities of one and the same transfer process sets $g = 1$; the Landauer bit-normalisation gives the same value independently. The uniqueness of both fixings is proved in Paper II[II].

The bound-state count established below (§3.3) does not depend on this last choice: it is the same across the whole family $-\frac{1}{2}\, d^2/d\phi^2 - g\, H(\sigma(\phi))$ for $g \in [0.80, 1.50]$ (verified pointwise; certificate included). The ε-dependent corrections of §5–§7 do rely on $g = 1$, since $\varepsilon = N - \sqrt{g}\, n_{\mathrm{WKB}}(1)$ changes sign inside the same window ($+0.51$ at $g = 0.80$, -0.41 at $g = 1.50$); at the lower edge the action count 2.49 dips below the Maslov threshold 2.5 while the exact count is unchanged—the threshold rule of §3.2 is a heuristic only.

Combining $V = -H$ as the potential with the kinetic term of the canonical normalisation ($g = 1$):

$$\left[-\frac{1}{2}\frac{d^2}{d\phi^2} - H(\sigma(\phi))\right]\Psi = E\,\Psi \tag{6}$$

This is nothing other than a Schrödinger equation: with only the three axioms of existence and the canonical normalisation $g = 1$ above as input, the eigenvalue equation of quantum mechanics emerges as the continuum limit of the transfer matrix of the canonical action.

The question posed at the head of this chapter—how many discrete states the well holds—is now well-posed, and §3.2–§3.3 answer it. Whatever that count turns out to be, it is a consequence of this construction—the coordinate ϕ fixed uniquely above together with a flat kinetic term. Writing the flat kinetic term in a different coordinate yields a different operator and a different spectrum, so the uniqueness of ϕ is essential.

The derivation chain so far is summarised as:

$$\text{A1–A3} \;\Rightarrow\; n^2{=}n,\ 0<\sigma<1 \;\Rightarrow\; Z = 1{+}e^{\phi} \xrightarrow{\text{Legendre}} V = -H$$
$$\xrightarrow{\text{transfer matrix}} \left[-\tfrac{1}{2}\partial_\phi^2 + V\right]\Psi = E\Psi$$

3.2 Three Bound States

Why do fermions come in exactly three generations—why τ, μ, e, and why not a fourth? This and the following subsection prove rigorously that the potential $V = -H$ has exactly three bound states and no fourth (hereafter $N = 3$), and identify them with the three generations of fermions. The identification is corroborated by the agreement of the mass ratios and mixing angles in §7.

The potential $V = -H$ is a well. It is deepest at the centre ($\phi = 0$) where $V = -\ln 2$, returning to zero at both ends $\phi \to \pm\infty$. In quantum mechanics, such a well confines discrete energy levels—by the same principle that the hydrogen atom confines electrons to discrete orbitals. In general, how many levels a well holds is set by two free parameters, its depth and its width—a deep, wide well holds many levels, a shallow, narrow well holds few.

The well of $V = -H$, however, has no such free parameters. Its depth is fixed as the number $\ln 2 \approx 0.693$ directly by the functional form of the Shannon entropy itself (§2.3), and the coefficient of the kinetic term that plays the role of width is likewise fixed to the unique value $g = 1$ (the uniqueness theorem of §3.1). In an ordinary well problem, depth and width are free to choose; here neither is—both are completely fixed by the form of the equations alone.

How many levels, then, does this uniquely fixed well hold? That can only be found by actually solving it and counting.

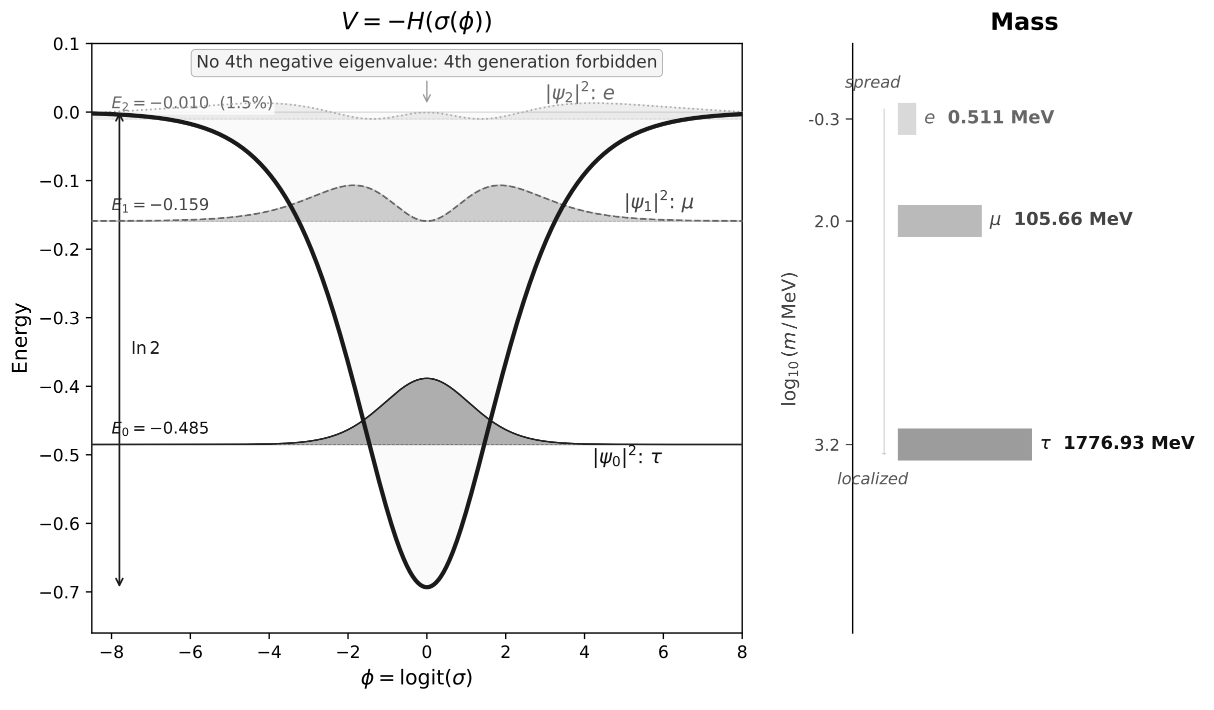

Figure 1: The potential $V = -H(\sigma(\phi))$ and the probability densities $|\psi_n|^2$ of its three bound states (left). More localised states correspond to heavier particles (right). No fourth negative eigenvalue: the prediction that no fourth sequential chiral generation exists.

We denote the number of levels confined in this well by N. A semiclassical (WKB) approximation estimates how many half-wavelengths fit inside the well. Each integer half-wavelength corresponds to one bound state:

$$n_{\mathrm{WKB}} = \frac{1}{\pi}\int_{-\infty}^{\infty} \sqrt{2H(\sigma(\phi))}\, d\phi = 2.781\,. \tag{7}$$

In the standard convention including the Maslov correction, the threshold for the k-th state is $n_{\mathrm{WKB}} > k - 1/2$. The value $n_{\mathrm{WKB}} = 2.781$ exceeds the third-state threshold 2.5 by 0.281, while falling short of the fourth-state threshold 3.5 by 0.719. The WKB estimate therefore suggests $N = 3$. We define $\varepsilon \equiv N - n_{\mathrm{WKB}} = 3 - 2.781 = 0.219$: the gap between the exact count and the semiclassical action count.[2] That $\varepsilon > 0$ says the third state is barely bound.[3] This ε is the source of every correction exponent in this paper: it controls the NLO corrections to the couplings and mixing angles of §5–§7. Nothing in this count is adjustable. The three ingredients of the construction are each unique: the coordinate and the normalisation are the unique ones of §3.1, and Shannon entropy is the unique entropy satisfying the Khinchin axioms, which include strong additivity (Khinchin's theorem [16, 17]). Under this construction, $N = 3$ is rigorously proved by interval arithmetic (§3.3).

3.3 Rigorous Proof

Theorem 1 ($N = 3$)**.** *The Schrödinger equation of §2 has exactly three bound states, and their eigenvalues are rigorously enclosed by interval arithmetic:*

$$\begin{aligned} E_0 &= -0.4850022531029642374452777658 \pm 10^{-24}, \\ E_1 &= -0.159052221618856177993419 0 \pm 10^{-24}, \\ E_2 &= -0.010423393062109459327616 \pm 10^{-22} \end{aligned}$$

No fourth negative eigenvalue exists; the essential spectrum begins at 0 (the small positive values seen on finite grids are discretisation artifacts of the box).

Proof (computer-assisted, interval arithmetic). The proof has three independent layers.

Counting. By parity the problem splits into a Neumann problem (even) and a Dirichlet problem (odd) on the half-line $[0, \infty)$. Propagating the Prüfer angle by interval enclosures (using the monotonicity of the potential), combined with analytic tail estimates, gives rigorous two-sided bounds on the angle at the trial energies. By Sturm oscillation theory the rotation of the angle counts the negative eigenvalues exactly: two in the even sector, one in the odd—$N = 3$.

Enclosures. The eigenvalue intervals above are certified by two-sided shooting with a verified high-order Taylor integrator over interval coefficients.

Independent checks. A separate implementation in Arb ball arithmetic re-certifies $N = 3$, and variational upper bounds from Rayleigh quotients of three sech-type (Pöschl–Teller-type [18]) trial functions confirm the existence of three negative eigenvalues. Three certificate scripts are included in the supplementary materials. ☐

[2] The rigorous interval enclosure 0.219188 is certified by the accompanying certificate script (rigorous Arb-ball integration with an analytic tail bound); the certified interval lies inside the certified interval of Paper II.

[3] Measured against the Maslov-corrected threshold, the same marginality reads $2.781 - 2.5 = 0.281$. We use ε because it compares two unambiguously defined quantities—the exact count and the action integral—whereas the threshold rule is an asymptotic heuristic; the $g = 0.80$ example of §3.1 shows it can fail at finite depth.

Identification with generations. Here we hypothesise that these three bound states ψ_0, ψ_1, ψ_2 are the mathematical structure of the three generations of fermions. The ordering then fixes which state is which generation: binding energy orders localisation ($|E_0| > |E_1| > |E_2|$, from most to least localised), and the generations carry the mass hierarchy—the most localised ψ_0 is the heaviest generation. For charged leptons, $\psi_0, \psi_1, \psi_2 \leftrightarrow \tau, \mu, e$.

At this stage the identification is provisional. In §7 the mass ratios and mixing angles are derived from it and agree with experiment; that agreement is what makes it credible. If the identification is correct, the absence of a fourth negative eigenvalue is the prediction that no fourth sequential chiral generation exists.

3.4 Three Spatial Dimensions

The preceding subsections proved $N = 3$. That proof lives in the internal space (the ϕ coordinate), and ϕ is not a spatial coordinate—as §8.2 shows, the internal and spacetime factors are independent. Under the provisional identification of §3.3, however, the fermions represented by the three bound states are particles that exist in real space. What, then, determines the dimension of that space? The answer is the consistency of realisation: the internal well must be realised, unbroken, on a particle in space. This section shows that the three conditions of this consistency — preservation of the well (the operator must not be deformed under the radial transfer, the A3-based criterion: $d = 1, 3$ only), spin-statistics ($d \geq 3$), and propagating gravity ($d \geq 3$) — have $d = 3$ as their unique common solution.

Consider the same well shape $V = -H$ realised as a radial profile in d-dimensional space. Here $\sigma(r)$ is the same well shape as $\sigma(\phi)$ on $\phi \in \mathbb{R}$, transferred to the radial variable $r \geq 0$; the difference in domain is addressed below in the comparison with full-line quantisation. Reducing the d-dimensional Schrödinger equation to a one-dimensional radial equation by standard partial-wave decomposition, an additional effective force $(d-1)(d-3)/(8r^2)$ appears:

$$V_{\text{eff}}(r) = -H(\sigma(r)) + \frac{(d-1)(d-3)}{8r^2}. \tag{8}$$

The numerator $(d-1)(d-3)$ vanishes for $d = 1$ and $d = 3$, leaving $V = -H$ unchanged. In other dimensions, this geometric term deforms the operator:

d	$(d-1)(d-3)/8$	Effect on the operator
1	0	No deformation
2	$-1/8$	Attractive term (tends to create extra states)
3	**0**	**No deformation: the well shape is preserved**
4	$+3/8$	Repulsive term (tends to remove the marginal third state)
≥ 5	≥ 1	Strong repulsion

Full-line quantisation ($\phi \in \mathbb{R}$) is essentially self-adjoint at the endpoints and requires no boundary condition. Under A3 (non-existence is not a state—no additional data is admissible at the boundary; the same conclusion as the exclusion of u-type coordinates in §3.1)[II], this canonical full-line quantisation is the reference, and a spatial realisation is required not to add an extra inverse-square term to the canonical operator. This requirement is met only when the coefficient $(d-1)(d-3)/8$ vanishes, i.e. only for $d = 1, 3$. Note that a vanishing inverse-square term does not make the full-line and radial problems identical—a half-line radial model adds endpoint structure at the origin (a choice of self-adjoint extension) and is a different model, outside the comparison fixed by this correspondence (Paper II).

Two realisation-consistency conditions exclude $d = 1$. First, the fermionic structure (CAR) is derived from A1–A3 (base-independent bivalence)$^{\mathrm{II}}$; the half-integer spin its spatial realisation requires holds by the spin-statistics theorem for $d \geq 3$ ($\pi_1(\mathrm{SO}(d)) = \mathbb{Z}_2$, giving the double cover $\mathrm{Spin}(d)$ with finite-dimensional spinor representations). Second, the realisation of a spin-2 field (gravity)—taken as a premise, as in §8.1, with its derivation left to a subsequent paper—propagating dynamically requires $d \geq 3$ (for $d \leq 2$ the Weyl tensor vanishes identically, leaving no local degrees of freedom).

We summarise this as a theorem.

Theorem 2 ($d = 3$)**.** *Under the above choice of full-line quantisation, $d = 3$ is the unique spatial dimension simultaneously satisfying:* (i) *the well operator is preserved with no extra geometric term: $d = 1, 3$;* (ii) *spin-statistics connection holds: $d \geq 3$;* (iii) *gravity propagates (Weyl tensor non-trivial): $d \geq 3$. Intersection: $\{3\}$.*

Proof. (i) centrifugal term $(d-1)(d-3)/(8r^2) = 0$ iff $d = 1$ or 3; (ii) $\pi_1(\mathrm{SO}(d)) = \mathbb{Z}_2$ for $d \geq 3$; (iii) the Weyl tensor vanishes identically in spacetime dimension ≤ 3, i.e. spatial $d \leq 2$; propagating gravity requires $d \geq 3$. □

4 Derivation of the Gauge Structure from the Projective Plane

The previous chapter derived two numbers from the Schrödinger equation: the bound-state count $N = 3$ (§3.3) and the spatial dimension $d = 3$ (§3.4). The two were derived independently—yet they take the same value, $N = d = 3$. On this coincidence, we construct the three-dimensional space $\mathbb{F}_3^3$ over the three-element field $\mathbb{F}_3$—why "3" becomes a field, and not a mere count, is shown in §4.1. This chapter shows that the geometric structure arising from it has the same mathematical construction as the Standard Model gauge structure.

4.1 Origin of the Gauge Structure

What gauge structure arises from $N = d = 3$? The key is degeneracy. A gauge symmetry requires degeneracy — the interchange of directions with the same energy.

On the Standard-Model side, quark colour (red, green, blue) is the archetype: every colour has the same energy, so the continuous rotation symmetry SU(3) stands. On the derived side, the generations — the three bound states of $V = -H$ — all have distinct energies ($E_0 \neq E_1 \neq E_2$, by the Sturm–Liouville theorem), so a preferred eigenbasis exists. The symmetry preserving this basis is discrete and cannot support a continuous gauge group.[4] The gauge directions therefore cannot arise from the interchange of generations — they must come from the geometry of the space $\mathbb{F}_3^3$, which we now construct.

[4] The contact index $b = 8/3$ in the mass formula of §7.1 is derived from the finite incidence geometry and is compatible with this (its numerical agreement with the SU(3) Casimir is an $N = 3$-only consistency check, §7.1).

$$N=3 \xrightarrow{\text{cycling}} C_3 \xrightarrow{\text{End}} \mathbb{F}_3 \xrightarrow{d=3} \mathbb{F}_3^3 \xrightarrow{\text{projectivisation}} \mathrm{PG}(2,\mathbb{F}_3)\,(13\text{ points}) \xrightarrow{\text{flag}} 9{+}3{+}1$$

From the coincidence $N = d = 3$, the three-dimensional space $\mathbb{F}_3^3$ over the field $\mathbb{F}_3 = \{0, 1, 2\}$ is determined. The origin of the field structure lies in the exchange of generations. The cyclic shifts of the three bound states ($1 \to 2 \to 3 \to 1$ and its iterates) form C_3, the unique normal subgroup of S_3, and carry any state to any other by exactly one shift. The maps preserving the composition of shifts are only three—multiplication by 0, by 1, and by 2 (the endomorphism ring)—and they form the field $\mathbb{F}_3 = \mathbb{Z}/3\mathbb{Z}$ with its addition and multiplication: here 3 becomes a field, not merely a count.

$$C_3 = \{e,\ c,\ c^2\}, \qquad \mathrm{End}(C_3) = \{\, c \mapsto c^k : k = 0, 1, 2 \,\} \cong \mathbb{Z}/3\mathbb{Z} = \mathbb{F}_3$$

Juxtaposing the three independent directions supplied by $d = 3$ over this $\mathbb{F}_3$ gives $\mathbb{F}_3^3$ ($|\mathbb{F}_3^3| = 3^3 = 27$ elements; the theorem-level construction is given in Paper II). The projective construction that follows requires a field (not merely a group); since $N = 3$ is prime, $\mathbb{F}_3 = \mathbb{Z}/3\mathbb{Z}$ is the unique field with N elements—indeed, the additive group of a three-element field has order 3, so it coincides with $\mathbb{Z}/3\mathbb{Z}$ generated by 1, and multiplication is then determined from addition by distributivity, making the operation tables unique.

Removing the origin and taking a two-dimensional cross-section gives $\mathbb{F}_3^2 = \{0, 1, 2\} \times \{0, 1, 2\}$—a 3×3 grid. These 9 points are the starting point for the construction.

Drawing lines on the 3×3 grid, lines with the same slope are parallel and do not intersect. In projective geometry, parallel lines meet at a single "point at infinity"—the same principle by which parallel railway tracks converge to a point on the horizon in perspective drawing. The lines on $\mathbb{F}_3^2$ have $|\mathbb{F}_3| + 1 = 4$ slopes ($0, 1, 2, \infty$—horizontal, right-diagonal, left-diagonal, vertical), each generating one point at infinity. These four points form the "line at infinity" L—the horizon.

Adding the 4 points at infinity to the original 9 gives $9 + 4 = 13$ points. This is the projective plane $\mathrm{PG}(2, \mathbb{F}_3)$—the space obtained by defining "points" as one-dimensional subspaces and "lines" as two-dimensional subspaces through the origin of $\mathbb{F}_3^3$, with a symmetric structure of 13 points, 13 lines, 4 points per line, and 4 lines through each point (Figure 2).

$$|\mathrm{PG}(2, \mathbb{F}_3)| = \frac{3^3 - 1}{3 - 1} = 13$$

Under the full symmetry $\mathrm{PGL}(3, \mathbb{F}_3)$ of $\mathrm{PG}(2, \mathbb{F}_3)$, all 13 points are equivalent and no partition arises. We now fix one of the 4 points on the line at infinity L as a reference point P_0—the pair (P_0, L) is called a flag. Fixing the flag is a choice of reference within the construction: $\mathrm{PGL}(3, \mathbb{F}_3)$ acts transitively on flags, so every choice yields an isomorphic partition. The fixing splits the 13 points into three orbits that do not mix under the transformations preserving the flag: 9 points off L (the 3×3 grid), 3 points on L excluding P_0, and P_0 itself. Choosing P_0 alone gives only 1+12 = two groups; choosing a flag produces 9+3+1. This is the minimal structure distinguishing three forces:

$$\underbrace{N^2}_{\text{9 colour}} + \underbrace{N}_{\text{3 weak}} + \underbrace{1}_{\text{1 hyp}} = N^2 + N + 1. \tag{9}$$

$N = d$ is satisfied only by $N = 3$ ($N = 2, 4, 5$ all have $d = 3 \neq N$). All numbers in the partition 9+3+1 are determined by N alone: $N^2 = 9$, $N{+}1 = 4$ (excluding the reference

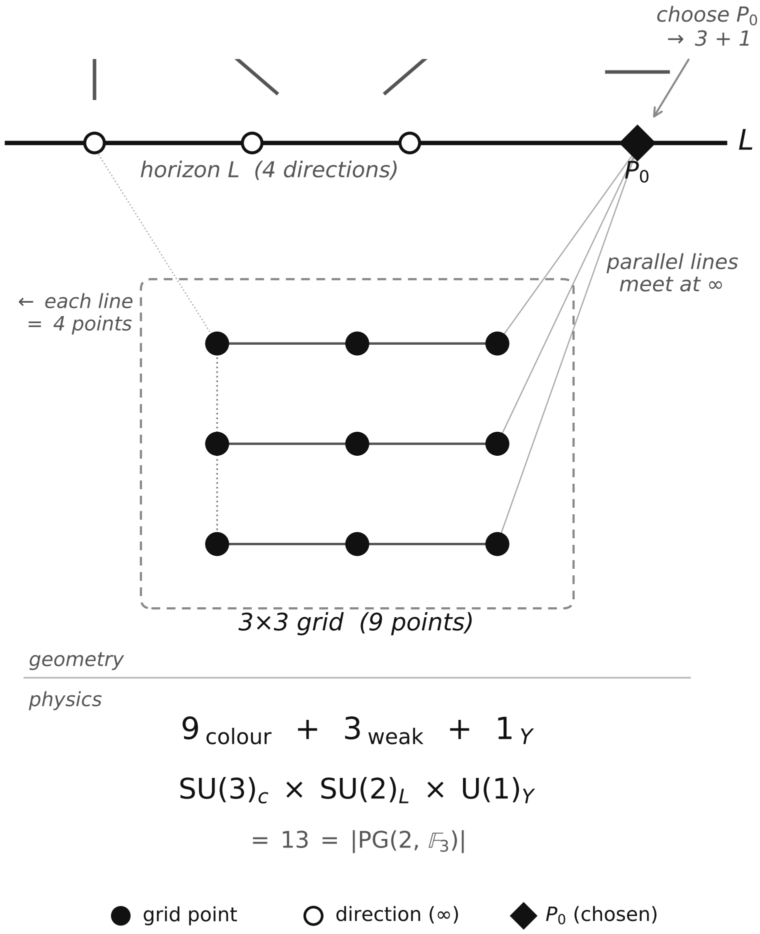

Figure 2: The projective plane $\mathrm{PG}(2, \mathbb{F}_3)$. Parallel lines on the 3×3 grid (9 points, filled circles) meet at directions on the line at infinity L (open circles). Choosing P_0 (filled diamond) produces the partition 9+3+1 = 13, corresponding to the sector structure of SU(3)×SU(2)×U(1).

point gives $N = 3$), reference point 1. Total: $N^2+N+1 = 13$. The 52 flags (point-line incidence pairs) provide the basis for the Laplacian (§5.3).

4.2 From 13 Points to the Standard Model Gauge Group

The partition 9+3+1 has the same form as the sector structure of the Standard Model gauge group $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)_Y$. Here we hypothesise that the three blocks of this partition are the gauge sectors of the Standard Model. At this stage this is a provisional identification; but the coupling constants (§5) and the mixing angles (§7) are derived from it and agree with experiment, which corroborates the identification.

9 points → SU(3). The 9 off-line points constitute the affine plane $\mathrm{AG}(2, \mathbb{F}_3)$. These 9 points are labelled by $(a, b) \in \mathbb{F}_3^2$ and naturally index the finite Weyl operators $W_{\{a,b\}} = X^a Z^b$ in the three-dimensional internal representation. The identity $W_{\{0,0\}}$ is the central component; the remaining 8 non-trivial Weyl components form the traceless sector. By the Killing–Cartan classification, the unique rank-2, dimension-8 compact simple Lie completion is $\mathfrak{su}(3)$. The 9th, central component corresponds to the global baryon number

$U(1)_B$ (an ungauged global charge; the exact anomaly-free combination is $B{-}L$, §7.4).
3 points → SU(2). The 3 points on L excluding P_0 give three non-trivial weak directions relative to the reference point. The minimal non-abelian compact Lie completion of these three directions is the 3-generator $\mathfrak{su}(2)$, whose doublet representation gives the weak SU(2) action.
1 point → U(1)$_Y$. The reference point P_0 gives the abelian central module phase, and physically corresponds to $U(1)_Y$.
Direct product guarantee. Translations of the affine plane fix the line at infinity L—shifting the grid does not change the "directions." The colour sector (9 points) transformations do not mix with the electroweak sector (3+1 points). This is a geometric property following from the incidence structure of $PG(2,\mathbb{F}_3)$: it yields a direct-product structure of three mutually non-mixing sectors—the same shape as the Standard-Model gauge group. The global form of the group (in the Standard Model, including the $\mathbb{Z}_6$ quotient [38]) is not fixed by the geometry alone; it is determined by the faithful action on the particle carrier (Paper II).

The complete derivation chain is:

$$\begin{aligned} n^2{=}n \xrightarrow{V=-H} N{=}3 \xrightarrow{d=3} \mathbb{F}_3^3 &\xrightarrow{\text{count directions}} 13 \text{ pts} \\ \xrightarrow{\text{flag}} 9{+}3{+}1 &\xrightarrow{\text{Killing–Cartan}} \mathfrak{su}(3)\oplus\mathfrak{su}(2)\oplus\mathfrak{u}(1). \end{aligned} \tag{10}$$

This mathematical result agrees with the experimentally established gauge structure $SU(3) \times SU(2) \times U(1)$.

The 4 on-line points correspond to the four real degrees of freedom of the SU(2) doublet — the three Goldstone bosons required for $SU(2){\times}U(1) \to U(1)_{\text{em}}$ breaking and the physical Higgs. $V = -H$ takes its minimum $-\ln 2$ at $\phi = 0$ ($\sigma = 1/2$) and returns to zero as $\phi \to \pm\infty$. This shape defines a stable well ($V''(0) = 1/4 > 0$). The vacuum sits at the interior point $\sigma = 1/2$ rather than at the vanishing-field points $\sigma \in \{0, 1\}$—the role played in the Standard Model by the negative quadratic term ($\mu^2 > 0$), moving the vacuum from the symmetric point to a finite field value, is here played by the shape of the well.

4.3 Origin of Fermion Diversity

§4.2 established that the 13 directions of $PG(2,\mathbb{F}_3)$ give a structure with the same shape as the Standard-Model gauge group. But the 13 directions do not correspond to fermions themselves. The 13 directions define the gauge symmetry — the structure of forces — and the fermion structure emerges as **representations** of this gauge group. Here we show why the single existence variable n gives rise to a diverse matter-particle structure.

Before the gauge structure emerges, n has no quantum numbers — no generation label, no colour, no isospin. Without distinguishing properties, multiple instances of n are identical (Leibniz's principle of the identity of indiscernibles), and n remains unique. The gauge structure of $PG(2,\mathbb{F}_3)$ changes this: the representations of the gauge group create diverse "slots" — configurations — in the internal space, with each configuration carrying an occupation number $n_i \in \{0, 1\}$. The configuration decomposition is:

$$\sum_i \hat{n}_i = \hat{n}, \qquad \hat{n}_i^2 = \hat{n}_i, \qquad \hat{n}_i\,\hat{n}_j = \delta_{ij}\,\hat{n}_i. \tag{11}$$

The first equation states that existence decomposes into configurations; the second that bivalence ($n^2{=}n$) is inherited by each configuration; the third that the unique unit of existence occupies exactly one configuration — the orthogonal resolution of the one-particle state space (the $\hat{n}_i$ are mutually orthogonal projections summing to $\hat{n}$).

This mutual exclusivity is not an independent assumption but a consequence of the bivalence of existence (A1) and its uniqueness (A3). The Pauli exclusion principle itself — no double occupancy of a single mode, together with exchange antisymmetry — is carried by the many-body realisation (the CAR construction)$^{\mathrm{II}}$, with each configuration as a mode. In the many-body realisation distinct configurations can be simultaneously occupied, but the occupation of each mode is restricted to $\{0, 1\}$ — this is the occupation-number representation of the Pauli exclusion principle [11]. $n_i = 0$ does not mean non-existence; it means existence occupies a different configuration.

Each configuration i is characterised by two labels: generation k (which bound state of $V = -H$, $k = 1, 2, 3$) and gauge representation α (which sector of $\mathrm{PG}(2, \mathbb{F}_3)$). That is, $n_i = n_{k,\alpha}$. The representations of SU(3)×SU(2)×U(1) give 16 states per generation. The chiral asymmetry (SU(2) acting only on left-handed fields) is part of this particle-species specification and is not derived in this paper. In Paper II, under the stated enumeration conditions, mixed configurations are excluded and the left/right assignment is unique up to a global orientation convention. The hypercharge values in the table below are the unique anomaly-free assignments compatible with the 9+3+1 sector structure, Yukawa closure, the inclusion of ν_R, and the specification of a single local abelian gauge factor ($B-L$ remains an ungauged global charge, §4.2); the operator-level derivation is likewise given in the same paper:

Representation	Content	States
$(\mathbf{3}, \mathbf{2}, 1/6)$	Left-handed quarks Q_L	6
$(\mathbf{3}, \mathbf{1}, 2/3)$	Right-handed up-type u_R	3
$(\mathbf{3}, \mathbf{1}, -1/3)$	Right-handed down-type d_R	3
$(\mathbf{1}, \mathbf{2}, -1/2)$	Left-handed leptons L_L	2
$(\mathbf{1}, \mathbf{1}, -1)$	Right-handed charged lepton e_R	1
$(\mathbf{1}, \mathbf{1}, 0)$	Right-handed neutrino ν_R	1

Total: 16×3 generations = 48 configurations. The inclusion of ν_R is required for Yukawa closure and for the $B-L$-protected Dirac-neutrino structure (see §7.4 of this paper). This makes the configuration decomposition concrete: the index i runs over the pair (k, α). The generators of SU(3)×SU(2)×U(1) correspond to 8+3+1 = 12 gauge bosons (8 gluons, 3 weak bosons W^+W^-Z, 1 photon). Together with the Higgs boson from §4.2, the 48 fermions, 12 gauge bosons, and 1 Higgs constitute the complete particle content of the Standard Model.

Thus, diverse matter (n_i) arises from unique existence (n). The key is that the gauge structure of $\mathrm{PG}(2, \mathbb{F}_3)$ creates "distinguishable slots" — without gauge symmetry, existence remains featureless and singular; with gauge symmetry, 48 species of fermions emerge. The diversity of matter is a consequence of gauge structure.

The internal consistency of this construction is independently confirmed by the eigenfunction participation ratio $D_{PR} = (\int \rho \, d\phi)^2 / \int \rho^2 \, d\phi$, with $\rho = \sum_n \psi_n^2$ the total density of the three bound states, $= 13.06 \approx |\mathrm{PG}|$ (reproduced by the verification code) and by the spectral decomposition of the 52×52 flag Laplacian.[5] No step in the derivation con-

[5]The finite-kernel theorem and its spectral gauge-decomposition corollary in Paper II.

tains a free parameter. The validity of this construction is supported by the fact that all independent physical quantities derived in §5–9 agree with experiment to 0.0006%–2.6% accuracy.

5 Gauge Coupling Constants

§4.2 derived the structure corresponding to the gauge group (which forces exist). This chapter derives their strengths (coupling constants).

5.1 Weinberg Angle

The partition 9+3+1 of §4.2 gives values corresponding to the coupling constants. The weight of each point is fixed by the symmetry prior to choosing a flag: $\mathrm{PGL}(3,\mathbb{F}_3)$ acts transitively on $\mathrm{PG}(2,3)$, so the 13 directions carry equal weight (invariant measure). Fixing the flag does not alter these weights; it only classifies the 13 points into 9+3+1. On this basis, the Weinberg angle [21] (weak mixing angle) $\sin^2\theta_W$ is read and identified as the fraction that the 3 weak directions occupy among all 13—taking all internal directions as the denominator[II]:

$$\sin^2\theta_W = \frac{N}{N^2+N+1} = \frac{3}{13} = 0.2308 \qquad (\text{expt: } 0.23122\ \overline{\text{MS}} \text{ at } M_Z,\ 0.2\%). \tag{12}$$

Alternatives such as taking only the four electroweak directions in the denominator correspond to a different observable type; that the type-preserving correspondence is unique to this fraction is proved in Paper II[II].

The tree-level prediction $3/13 = 0.23077$ deviates from the $\overline{\text{MS}}$-scheme experimental value 0.23122 ± 0.00003 [22] by 0.2%, consistent with one-loop radiative corrections. This angle gives the tree-level W/Z mass ratio: $m_W/m_Z = \cos\theta_W = \sqrt{10/13} = 0.8771$ (expt: 0.8813, 0.5%).

5.2 Strong Coupling Constant

The strong coupling constant follows by distributing the weak-sector fraction over the rank of the colour sector. Of the 9 colour-sector directions, the independent directions number $\mathrm{rank}(\mathrm{SU}(3)) = 2$ (Cartan generators); equidistributing $\sin^2\theta_W$ over these 2 directions: $\alpha_s^{(0)} = \sin^2\theta_W/(N-1) = 3/26 = 0.1154$ (the uniqueness of this distribution rule is proven[II]). The spectral marginality $\varepsilon = 0.219$ from §3 provides the NLO correction. The correction is distributed over $N^2+1 = 10$ non-weak PG directions (colour 9 + hypercharge 1):

$$\alpha_s = \frac{3}{26}\left(1 + \frac{\varepsilon}{N^2+1}\right) = 0.1179 \qquad (\text{expt: } 0.1180\ \overline{\text{MS}} \text{ at } M_Z,\ 0.07\%). \tag{13}$$

The value $k = N^2+1 = 10$ follows from the number of non-weak directions in $\mathrm{PG}(2,\mathbb{F}_3)$. Throughout this paper the NLO corrections take the form $1\pm\ell\varepsilon/k$, where the denominator k is the size of the finite support over which the correction is distributed—$k = N^2+1 = 10$ for the non-weak sector (α_s; affine + hypercharge), $k = N^2 = 9$ for generation mixing ($\sin\theta_C$; affine plane), $k = |\mathrm{PG}| = 13$ for the full set of PG modes (m_H/v, $\sin^2\theta_{23}$; full projective space), and $k = N = 3$ for the three weak directions ($\sin^2\theta_{13}$). The assignments of (ℓ, k) to each observable, including the sign and coefficient ℓ, are each proven unique[II].

5.3 Fine-Structure Constant

The Weinberg angle and strong coupling constant were determined by simple point ratios on $PG(2, 3)$. The fine-structure constant requires more—the full spectral structure of the internal space.

The inverse electromagnetic coupling $1/\alpha$ measures the weakness of the electromagnetic interaction. The observed charge is the charge screened by the matter fields—the stronger the screening, the weaker the coupling and the larger $1/\alpha$. In this framework $1/\alpha$ is given (up to the self-consistent correction $-\alpha$) by the sum of the bare photon propagation cost λ_1 and the matter screening quantity Z, and the Thomson-point value is dominated by the screening term.

In 4D QED this structure is formalised as the Dyson equation. In 0D the same structure holds but becomes algebraically exact. The propagator is a constant, so the vertex correction and self-energy are determined by the same resolvent, and the 4D perturbative cancellation $Z_1 = Z_2$ holds as an exact algebraic identity (proof in Paper II). The self-consistency equation is:

$$\frac{1}{\alpha} + \alpha = \lambda_1 + Z. \tag{14}$$

The right-hand side has two terms: λ_1 is the bare photon inverse propagator (propagation cost) and Z corresponds to the scalar-channel spectral quantity (screening). This correspondence is a spectral one—Z is the resolvent trace whose energy-denominator structure matches the matter vacuum polarisation.

The form of the equation (the left-hand side $1/\alpha+\alpha$ with unit coefficient) is derived[II]—what remains external is the comparison convention identifying the incidence current with the Thomson-point electromagnetic current. The full justification of this correspondence as a self-consistent prediction, and the proof that the physical root is unique under this identification, are both given in the same paper. The dominant contribution to Z is $1/|E_2| = 95.94$; since E_2 is rigorously certified by the interval arithmetic of §3.3, this sensitivity introduces no uncertainty.

The minimum cost λ_1 for gauge-field propagation through internal space is the smallest non-zero eigenvalue of the Laplacian on internal space. Since gauge interactions involve both positions (points) and directions (lines), the natural object for diffusion is a flag (point-line incidence pair). The spectrum of the graph Laplacian $\mathcal{L}$ on the flag graph of $PG(2, \mathbb{F}_3)$, a 52×52 matrix, is:

Eigenvalue	Value at $N = 3$	Multiplicity
0	0	1
$(N+1) - \sqrt{N}$	2.268	12
$(N+1) + \sqrt{N}$	5.732	12
$2(N+1)$	8	27

This spectrum follows elementarily from the incidence counts of the projective plane alone. Each point lies on 4 lines and each line carries 4 points, so flag adjacency (sharing the point or sharing the line) forms a 6-regular graph, and the Laplacian can be written

$$\mathcal{L} = 8\,\mathbf{1} - P - \Lambda$$

where P (Λ) is the operator summing over the 4 flags with the same point (the same line). On constant functions $P = \Lambda = 4$, so $\mathcal{L} = 0$ (multiplicity 1). On the space of functions whose sum vanishes within every point group and within every line group—of dimension $52-13-13+1=27$—one has $P=\Lambda=0$, so $\mathcal{L}=8$ (multiplicity 27).

The remaining 24 dimensions are spanned by pairs of a zero-sum point function u and a zero-sum line function v, on which $\mathcal{L}$ acts as $(u,v)\mapsto(4u-Mv,\,4v-M^{\mathsf{T}}u)$, with M the 13×13 point-line incidence matrix. Since exactly one line passes through two distinct points, $MM^{\mathsf{T}}=3I+J$ (J the all-ones matrix), which on zero-sum functions reduces to $MM^{\mathsf{T}}=3I$. Hence $(4-\lambda)^2=3$, i.e. $\lambda=4\mp\sqrt{3}$ (multiplicity 12 each). The $\sqrt{3}$ comes from the uniqueness of the line joining two points; the $4=N+1$ from the number of lines through a point.

The smallest non-zero eigenvalue $\lambda_1=4-\sqrt{3}=2.268$ is the quantity that corresponds to the bare photon inverse propagator in the self-consistent equation $1/\alpha+\alpha=\lambda_1+Z$ above. $\lambda_1>0$ is consistent with the confinement picture—the positive spectral gap corresponds to colour-charged states not propagating freely in the asymptotic regime.

The screening term Z corresponds to the scalar-channel spectral quantity from three generations of matter fields, determined by the resolvent of the internal kinetic operator. The internal kinetic operator is a tensor sum of the generation sector (K_{gen}, whose eigenvalues are the binding energies $|E_n|$) and the gauge sector ($\mathcal{L}$, eigenvalues λ_k):

$$K_{\text{int}}=K_{\text{gen}}\otimes\mathbf{1}+\mathbf{1}\otimes\mathcal{L}$$

This form—the tensor sum, with both factors measured in the same unit at relative scale 1—is established as a uniqueness theorem[II] under the single-clock unit clause (product couplings, twisted products, and rescalings of the relative unit are excluded). Decomposing the trace of the resolvent by multiplicity:

$$Z=\mathrm{Tr}[1/K_{\text{int}}]=\zeta_V(1)+12\,S_1+12\,S_2+27\,S_3=134.77613$$

where $\zeta_V(1)=\sum_n|E_n|^{-1}$ and $S_i=\sum_n(|E_n|+\lambda_i)^{-1}$. The intermediate values are: $\zeta_V(1)=104.28714$ (dominated by $E_2^{-1}=95.94$), $12S_1=14.57025$, $12S_2=6.05684$, $27S_3=9.86190$. The 27-dimensional sector (multiplicity $27=3^3$) contributes $9.86\approx N^2$, providing the spectral structure corresponding to the QCD vacuum-polarisation correction (all intermediate values are reproducible with the accompanying verification code).

Substituting $\lambda_1=4-\sqrt{3}=2.26795$ and $Z=134.77613$ into the self-consistency equation:

$$\frac{1}{\alpha}=\frac{(Z{+}\lambda_1)+\sqrt{(Z{+}\lambda_1)^2-4}}{2}=137.0368\quad(\text{expt: }137.036,\ 0.0006\%).\tag{15}$$

Adding the correction terms (the proof is given in Paper II):

$$\frac{1}{\alpha}+\alpha=\Lambda_0-x\Big(1+\frac{\alpha}{2}\Big)+\frac{3}{4}\varepsilon^2x+\frac{1589}{5408}x^2-\frac{2}{13}\frac{x^3}{1+x},\qquad x=\frac{9\alpha}{26\pi}$$

where $\Lambda_0=\lambda_1+Z$ is the right-hand side of the leading term and x is the same expansion parameter as the colour screening of §7.3. The origin of each term—the self-consistent screening correction, the generation branching for ε^2x, the second-order screening count for x^2, and the first return of the 52-flag orbit for x^3—and the proof that the series closes in this form are given in Paper II. The physical root of this equation is

$$\frac{1}{\alpha}\bigg|_{\text{all orders}}=137.035999084$$

The value is certified to 13 digits by the interval [137.0359990843979, 137.0359990844119] (Paper II). We note that the experimental recommendations split by measurement system: CODATA 2018, whose principal input is the Cs photon-recoil measurement, gives 137.035999084(21) — agreement to all displayed digits — while CODATA 2022, based on the Rb recoil, gives 137.035999177(21), 4.4σ away. The Cs and Rb measurements themselves disagree by more than 5σ; on this open experimental dispute the framework gives the prediction that the fine-structure constant lies on the Cs side (Table 4).

6 Higgs Sector and Electroweak Symmetry Breaking

§5 derived the gauge coupling constants. The Standard Model has one more free parameter in the scalar sector — the Higgs self-coupling λ. This chapter derives the mathematical structure corresponding to it and shows the pinning of the internal vacuum and the boundedness of the internal potential (no instability channel).

6.1 Symmetry Breaking and Higgs Mass

Electroweak symmetry breaking is the statement that, while the dynamical laws remain symmetric, the vacuum (ground state) departs from the symmetric point. Since the W and Z are massive while the photon is massless, the vacuum cannot sit at the field-vanishing point (the symmetric point)—breaking is required by experiment. The Standard Model realises this through the assumption $\mu^2 > 0$ in the Higgs potential $V = -\mu^2|\Phi|^2 + \lambda|\Phi|^4$: it destabilises the origin, rolling the field to a finite value. But "why $\mu^2 > 0$" is not explained within the Standard Model.

The $V = -H(\sigma(\phi))$ of this paper answers this question through structure. The binary Shannon entropy H vanishes at both endpoints $\sigma \in \{0, 1\}$ (the trivial configurations corresponding to the field-vanishing point) and is maximal at the interior point $\sigma = 1/2$, so $V = -H$ is a stable well that is 0 at the endpoints and reaches its minimum $-\ln 2$ at the interior point ($V''(0) = 1/4 > 0$). Moreover A2 ($\sigma \neq 1$) and A3 ($\sigma \neq 0$) forbid the endpoints themselves. The ground state therefore necessarily sits at the interior point—the condition that the vacuum cannot sit at the symmetric point, i.e. the condition for breaking, follows from the shape of the entropy and the axioms rather than from destabilising the origin. What the Standard Model assumes as $\mu^2 > 0$ becomes here a consequence of the shape of V. The Higgs-mass formula uses the positive curvature $V''(0) = 1/4$ at this minimum.

Under the identification of §4.2, the line structure of $\mathrm{PG}(2, \mathbb{F}_3)$ corresponds to the Higgs SU(2) doublet. The curvature at the centre—the Fisher metric at $\sigma = 1/2$—gives the Higgs mass:

$$V''(0) = g_F(0) = \sigma(1-\sigma)\big|_{\sigma=1/2} = \tfrac{1}{4}$$

$$m_H/v = \sqrt{V''(0)} = 1/2 \quad \text{(LO)}. \tag{16}$$

The NLO correction enters from all $|\mathrm{PG}| = 13$ modes of the Laplacian matrix $\mathcal{L}$:

$$m_H/v = \frac{1}{2}\sqrt{1 + 2\varepsilon/|\mathrm{PG}|} = 0.5084 \quad \text{(experiment: } 0.5082,\ 0.03\%\text{)}. \tag{17}$$

This corresponds to

$$\lambda = \tfrac{1}{2}(m_H/v)^2 = 0.1292 \quad \text{(expt: } 0.1291,\ 0.06\%\text{)}$$

6.2 Vacuum Stability and the Hierarchy Problem

The Standard Model Higgs potential $V_{\mathrm{SM}} = -\mu^2 h^2 + \lambda h^4$ diverges as $h \to \infty$, and the UV correction $\delta\mu^2 \propto \Lambda^2$ requires $\sim 10^{34}$ fine-tuning between the electroweak and Planck scales (hierarchy problem). Furthermore, the Higgs quartic coupling $\lambda(\mu)$ runs negative at $\mu \sim 10^{10}$ GeV, making the electroweak vacuum cosmologically metastable.

In the internal sector of $V = -H$, the instabilities corresponding to these problems do not arise. $V = -H$ is bounded ($V \in [-\ln 2, 0]$), with no high-energy divergence. The well depth $|V_{\min}| = \ln 2$ and curvature $V''(0) = 1/4$ are both $O(1)$ algebraic constants; their ratio gives $|V_{\min}|/V''(0) = 4\ln 2 \approx 2.77$, a ratio of internal quantities, an order-unity number requiring no fine-tuning:

$$V \in [-\ln 2,\ 0], \qquad |V_{\min}|/V''(0) = 4\ln 2 \approx 2.77$$

Relating this O(1) ratio to the 4D v^2/Λ^2, and interpreting the boundedness as the reason the running coupling $\lambda(\mu)$ in 4D does not turn negative, both require the reconstruction matching (likewise the full 4D vacuum-stability interpretation and the correspondence to thermal properties such as symmetry restoration at high temperature)[II]. The shape of V is fixed by the axioms, and scale dependence does not alter the algebraic structure of V. Furthermore, $V = -H$ is uniquely determined as the canonical free energy of $n \in \{0, 1\}$ (§2). Within the minimal operator algebra of the theory, no second fundamental scalar potential is generated. The minimal reconstruction thus predicts a single Higgs doublet.

6.3 Radiative Corrections and Summary of Couplings

Radiative corrections. The predictions above are algebraic values, and their comparison with M_Z-scale experiments requires the 4D RG evolution and loop corrections of the running quantities (§8.3). The next correction to the weak angle—the radiative correction, with expansion parameter α, that carries the tree value 3/13 to the $\overline{\mathrm{MS}}$ value at M_Z—is fixed in closed form by the framework itself:

$$\Delta_{\mathrm{rad}} = \alpha \cdot \frac{5N^3}{|\mathrm{PG}|^3} = \alpha \cdot \frac{135}{2197} = 0.00045$$

(N and $|\mathrm{PG}|$ are quantities of this paper; the uniqueness of this product form, including the coefficient 5, is proved as a theorem[II].) This gives $\sin^2\theta_W = 3/13 + \Delta_{\mathrm{rad}} = 0.23077 + 0.00045 = 0.23122$, in agreement with the $\overline{\mathrm{MS}}$ value 0.23122 ± 0.00003 [22] within errors. Similarly, $m_W/m_Z = \sqrt{10/13} = 0.8771$ is 0.5% below experiment; this residual has approximately the size and sign of the standard SM top-quark $\Delta\rho$ correction. That the custodial response is carried by exactly one neutral component of the one Higgs doublet is a theorem; what remains as identification is the naming of the experimental definition points (the on-shell convention) and the readout word. Under that identification, $m_W/m_Z = 0.88137$ follows uniquely and agrees with experiment at $+1.0\sigma$[II].

Summary. $\sin^2\theta_W(M_Z) = 0.23122$ ($< 0.01\%$), $\alpha_s(M_Z) = 0.1179$ (0.07%), $1/\alpha_{\mathrm{em}} = 137.0368$ (0.0006%; 137.035999084 at all orders[II]), $m_H/v = 0.5084$ (0.03%). All four coupling constants are derived from $V = -H$ and PG(2, 3) and agree with experiment. The vacuum is pinned at an interior point, the internal potential is bounded (no instability channel), and there is only one Higgs. The correspondence to 4D vacuum stability and the hierarchy problem is a matter of reconstruction matching.

7 Fermion Masses and Mixing

§4–6 established the structures corresponding to gauge structure, coupling constants, and the Higgs mass. Based on the provisional identification of the generations made in §3, this chapter derives formulas for the mass ratios and mixing angles from the localisation of the bound states. The derived values agree to high precision with the mass ratios of charged leptons, quarks, and neutrinos, and with the CKM/PMNS mixing angles — this agreement corroborates the provisional identification of §3.

7.1 Mass Formula

The three bound states of $V = -H$ differ in localisation. The deepest ψ_0 is localised at the well bottom and couples most strongly to the gauge field, corresponding to the heaviest τ:

Bound state	Energy	Localisation	Particle	Mass (MeV)
ψ_0	$E_0 = -0.485$	High (well bottom)	τ	1777
ψ_1	$E_1 = -0.159$	Medium	μ	106
ψ_2	$E_2 = -0.010$	Low (well edge)	e	0.511

The mass ratios are determined by the localisation of the wave functions, not as independent parameters. The hierarchy that the Standard Model inputs by hand as three Yukawa couplings is given here by the difference in localisation of the three modes of one well. The key quantity is $R = \ln(m_\tau/m_e)/\ln(m_\mu/m_e)$, which compresses the three-body mass hierarchy into a single number.

$$V = -H \xrightarrow{\S 3} \psi_n \xrightarrow{\text{IPR}} \text{localisation} \xrightarrow{b} m_n \to R = 1.5293 \tag{18}$$

Each eigenstate has a binding energy $|E_n|$ and a localisation, measured by the inverse participation ratio $\mathrm{IPR}_n = \int |\psi_n|^4 \, d\phi$. Since the coupling to the gauge field is proportional to the wave-function amplitude at the interaction point, a more localised state acquires a larger mass: $m_n \propto \mathrm{IPR}_n^b$.

The exponent b is not a free parameter. In 0D there are no loop integrals, hence no running that would generate an anomalous dimension dynamically. By the Peter–Weyl theorem for finite groups, the heat kernel on the group is the finite sum $K(t) = \sum_\rho \dim(\rho)\, \chi_\rho(g)\, e^{-\lambda_\rho t}$, admitting only exponential decay with no power-law terms—the exponent is a structurally fixed quantity. The exponent b is the contact index derived from the normalisation of the finite incidence geometry $\mathrm{PG}(2, \mathbb{F}_3)$: with the normalised single-leg index $w_{\text{leg}} = (N{+}1)/N$, $b = 2w_{\text{leg}} = 8/3$[6]. The uniqueness of this mapping rule is a theorem of Paper II[II7].

To this, a small correction from the anharmonicity of $V = -H$ is added. The virial ratio $|\langle V\rangle_n|/T_n$ (potential to kinetic energy) measures how differently each mode samples the well, and its exponent is the per-mode marginality $\varepsilon/N = 0.219/3 = 0.073$ (ε from §3.2).

[6] This is not the physical colour charge of leptons. The exponent b is the contact index of the incidence geometry; its numerical agreement with the SU(3) fundamental Casimir, $2C_F = 8/3$, holds only at $N = 3$—$(N^2{-}1)/2N = (N{+}1)/N$ only for $N = 3$—and is a consistency check at $N = 3$. See Paper II [20].

[7] The uniqueness of this specification is proven in Paper II: both the functional form of the mass response and the exponents ($b = 2w_{\text{leg}} = 8/3$ and $\varepsilon/3$) are fixed by uniqueness theorems.

The complete mass formula is:

$$m_n \propto |E_n| \times \mathrm{IPR}_n^b \times \left(\frac{|\langle V\rangle_n|}{T_n}\right)^{\varepsilon/N}. \tag{19}$$

The three factors have different origins:

Factor	Physical meaning	Determined by
$\lvert E_n\rvert$	Energy scale	Binding energy
IPR^b	Localisation $\to$ coupling strength	Finite incidence geometry (contact index)
$(\lvert\langle V\rangle\rvert/T)^{\varepsilon/N}$	Anharmonic correction	Spectral marginality

All exponents are determined by $V = -H$. The result:[8]

$$R = \frac{\ln(m_\tau/m_e)}{\ln(m_\mu/m_e)} = 1.5293 \qquad \text{(experiment: } 1.5294,\ 0.006\%\text{)}. \tag{20}$$

Koide's relation $Q \approx 3/2$ [23] (this paper's convention $Q = (\sum\sqrt{m})^2/\sum m$ is the reciprocal of the standard 2/3) is not an input that determines b, but a cross-check—a verification on the output side. Parametrising the mass formula in Koide form yields the pair (b, c), where c is the exponent of the virial correction. Back-solving, for each b, the c required to force $Q = 3/2$, only $b = 2w_{\mathrm{leg}} = 8/3$ returns a value, 0.0730, matching the spectral value $\varepsilon/N = 0.073$ (to 0.07%):

b	$c\,(Q{=}3/2)$	Deviation
2.0	0.437	500%
2.5	0.163	120%
$\mathbf{8/3 = 2w_{\mathrm{leg}}}$	**0.0730**	**0.07%**
2.78	0.012	84%
3.0	−0.107	246%

The individual mass ratios:

	Prediction	Experiment	Accuracy
m_τ/m_e	3481	3477	0.1%
m_μ/m_e	207.0	206.8	0.1%
R	1.5293	1.5294	0.006%
Q (Koide)	1.499985	1.500005(11)	2σ (from PDG masses)

The three lepton masses are not independent parameters but three modes of a single well. The 0.006% agreement of $R = 1.5293$ quantitatively supports the identification of the bound states with the lepton generations. The stability of the Koide relation under radiative corrections is discussed by Sumino [43].

[8] All spectral quantities are computed with $|\phi|_{\max} = 100$, $\Delta\phi = 1.25\times10^{-4}$ ($N_{\mathrm{grid}} = 1600001$), verified stable under doubling of grid resolution and box enlargement.

7.2 Quark Masses

The mass formula of §7.1 reproduced $R = 1.5293$ for leptons. Can the same formula be applied to quarks? The formula itself does not change—only the well depth changes.

Leptons are colour singlets and feel $V = -H$ as a single copy. Quarks carry three colours. The colour-triplet well composes the three colour contributions—isomorphic copies of the same Bernoulli degree of freedom—as independent factors pulled back to the common existence coordinate ϕ: $Z_3 = Z_1^3$, additivity of the logarithm gives $\log Z_3 = 3 \log Z_1$, and additivity of the Legendre dual gives $V = -3H^{\mathrm{II}}$; weak isospin does not deepen the well, because the doublet is an orientation structure within a single copy—it fixes which component is called $I_3 = +1/2$ (the g_c correction below)—and adds no second copy. The deeper well supports 5 bound states ($n_{\mathrm{WKB}} = 4.82$). In the tables below, κ denotes the well-depth multiplier (the κ in $V = -\kappa H$):

Sector	Operator	κ	Bound states	Modes used
Lepton	$-H$	1	3	$(0, 1, 2) = \tau, \mu, e$
Up-type	$-3H$	3	5	$(1, 2, 3) = t, c, u$
Down-type	$-3H$ (corr. in kinetic term)	3	4	$(0, 1, 2) = b, s, d$

Up-type. Mode 0 is deeply confined and belongs to the $I_3 = -1/2$ sector. Up-type quarks ($I_3 = +1/2$) use modes $(1, 2, 3)$, giving $R_{\mathrm{up}} = 1.777$ (experiment: 1.770, 0.4%).
Down-type. Up-type and down-type belong to the same SU(2) doublet and feel the same $V = -3H$, but differ in the sign of weak isospin ($I_3 = +1/2$ vs $-1/2$). This sign supplies a correction to the mass formula.

The total gauge charge of the SU(2) doublet (u, d), $2C_F + 1/N_c = N_c$, is an exact algebraic identity. The physical mechanism originates from $\mathrm{PG}(2, \mathbb{F}_3)$ geometry: among the $N{+}1 = 4$ points on line L, the reference point P_0 is the reference that orients the doublet (which component is called $I_3 = +1/2$), and the remaining 3 points generate SU(2). Reversing this orientation flips the sign of I_3 and reverses the SU(2) action, coupling the total charge N_c to $g_F(\phi) = \sigma(1{-}\sigma)$ (the Bernoulli Fisher metric $g_F = \mathrm{Var}(n)$ of §2.2; its uniqueness is Čencov's theorem [13]) with a negative sign. The effective mass is:

$$m_{\mathrm{eff}}(\phi) = 1 - N_c\, \sigma(1{-}\sigma) \tag{21}$$

This is the static Fisher-limit form of the down-sector mass formula. The operator actually solved is the symmetric position-dependent-mass form $-\frac{1}{2}\, \partial_\phi\, (1/m_{\mathrm{eff}})\, \partial_\phi - 3H$ (identical to the discretisation in the accompanying code).

The correction coefficient $g_c = -N_c = -3$ (distinct from the normalisation g of §3.1) is fixed by the algebraic structure of the SU(2) doublet. The corrected operator has 4 negative eigenvalues (-1.5709, -0.5455, -0.1952, -0.00733; stable for box size $L_{\mathrm{box}} = 40/80/120$), of which 3 modes $(0, 1, 2)$ correspond to the physical window b, s, d—the physical window is uniquely determined by type exclusivity and order preservation$^{\mathrm{II}}$ (the window differs sector by sector: all 3 modes for leptons, $(1, 2, 3)$ for up-type, $(0, 1, 2)$ for down-type). With modes $(0, 1, 2)$: $R_{\mathrm{down}} = 2.273$ (experiment: 2.276, 0.1%).

The mass ratios of all three charged-fermion sectors are reproduced:

Sector	κ	g_c	R (pred.)	R (expt.)	Accuracy
Lepton	1	0	1.529	1.529	0.006%
Up	$N_c = 3$	0	1.777	1.770	0.4%
Down	$N_c = 3$	$-N_c$	2.273	2.276	0.1%

Here R captures the shape of the log-hierarchy—it is invariant under a uniform stretching of the mass ratios (raising every m_i/m_j to a common power)—not the individual masses. The accuracy column gives the deviation of the prediction from the experimental central value. The experimental R inherits the light-quark mass uncertainties (PDG 2026: $m_u = 2.16 \pm 0.07$ MeV, etc.), which propagate to a width of about ± 0.15–0.25%, and the comparison further depends on the scheme and scale conventions for each mass (light quarks $\overline{\text{MS}}$ at 2 GeV, c and b at $m(m)$, t from direct measurement).[9]

7.3 CKM Mixing

§7.1–7.2 established the mass ratios. With masses determined, we now derive the inter-generation mixing [24]. Why is V_{ub} small, and why does the Cabibbo angle [19] take its particular value? The Standard Model does not explain these. In $V = -H$, the potential symmetry corresponds to the former and the marginality of the third generation to the latter.

The potential $V = -H$ is symmetric: $V(\phi) = V(-\phi)$. This fixes the parity of each eigenfunction: ψ_0 even, ψ_1 odd, ψ_2 even. The direct overlap between distinct modes vanishes by Sturm–Liouville orthogonality: $\langle\psi_0|\psi_2\rangle = 0$. Since ψ_0 and ψ_2 are both even, $\langle\psi_0|M|\psi_2\rangle = 0$ for a V-parity-odd mixing operator M—the leading $0 \leftrightarrow 2$ transition is forbidden:

$$V_{ub}\big|_{\text{tree}} = 0 \tag{22}$$

The measured value $|V_{ub}| \approx 0.004$ arises at higher orders of the Wolfenstein hierarchy [25]. V-parity makes the tree/direct term exactly zero and suppresses distant-generation transitions to at least $O(\varepsilon^3)$; the first non-zero term is given by the closed form below.

The Cabibbo angle follows from spectral marginality. The third bound state ψ_2 is marginally bound—its binding energy $|E_2| = 0.010$ is only 1.5% of the well depth. The parameter measuring this marginality is the gap between the exact bound-state count and the semiclassical action count: $\varepsilon = N - n_{\text{WKB}} = 3 - 2.781 = 0.219$.

Mixing angles reflect the mismatch between mass and weak eigenstates; the marginality of the third bound state ($\varepsilon > 0$) is the source of mixing, and $\varepsilon \to 0$ removes it, leaving the three generations intact but with vanishing flavour transitions. The expression giving the angle as a series in ε (which operator carries the transition, and what fixes its order and denominator) is proven unique per observable[II] (see the list in §5.2). To second order:

$$\sin\theta_C = \varepsilon\Big(1 + \frac{\varepsilon}{N^2}\Big) = 0.2245 \qquad \text{(experiment: } 0.2243,\ 0.10\%\text{).} \tag{23}$$

This value is the unscreened internal value. The value composed with colour screening is $\sin\theta_C \cdot (1 - x) = 0.22435$ ($x = N^2\alpha/(2\pi|\text{PG}|) = 9\alpha/(26\pi)$ is the colour-screening expansion parameter; derived in Paper II).

The Cabibbo angle is a direct expression of the fact that the third generation barely exists: the generation-sector spectral quantity $\varepsilon = 0.219$, defined in §3 from $V = -H$ alone and entering all mixing-angle expressions in common. If the third bound state were slightly more deeply bound ($\varepsilon \to 0$), inter-generation mixing would vanish and all flavour transitions would be forbidden.

[9] The individual quark magnitudes are not derived in this series at present: Paper II derives the within-sector ratios and shows that the readout normalisation of the quark sectors is not fixed by the internal structure alone.

The Wolfenstein hierarchy follows from the Cabibbo angle [25]: $|V_{us}| \sim \varepsilon$, $|V_{cb}| \sim \varepsilon^2$, and $|V_{ub}|$ is suppressed to at least $O(\varepsilon^3)$ with first non-zero term $2\varepsilon^4$ (experiment: 0.225, 0.041, 0.004). NLO corrections take the form $1\pm\ell\varepsilon/k$. The assignment of the denominator k and of the sign and coefficient ℓ is as listed in §5.2 ($1+\varepsilon/9$ for $\sin\theta_C$, $1-2\varepsilon/3$ for $\sin^2\theta_{13}$). The full closed forms, proved as a theorem in Paper II, are

$$|V_{cb}| = \varepsilon^2\left(1-\frac{2\varepsilon}{3}\right) = 0.0410, \qquad |V_{ub}| = 2\varepsilon^4\left(1-\frac{2\varepsilon}{3}\right)\left(1-\frac{3\varepsilon}{26}\right) = 0.00384 \tag{24}$$

(experiment: 0.041, 0.004).

The origin of these exponents and of the coefficient 2 is discrete. Generation transitions are carried only by parity-odd permutations of the three generations (a consequence of V-parity). The adjacent transitions $1 \leftrightarrow 2$ and $2 \leftrightarrow 3$ are realised by the transpositions t_{12} and t_{23}. With respect to the flag $V_1 \subset V_2$ fixed by the generation order (the line spanned by the first generation and the plane spanned by the first two), count the cost of a wall as the shallowest flag level it moves: t_{12} moves V_1, cost 1; t_{23} fixes V_1 and moves only V_2, cost 2—these are the exponents of $|V_{us}| \sim \varepsilon$ and $|V_{cb}| \sim \varepsilon^2$.

Parity makes the direct one-step $1 \leftrightarrow 3$ element vanish, but it does not by itself forbid the two-step composite $1 \to 2 \to 3$: two odd insertions compose to an even one, and the product of the two matrix elements is nonzero. What separates them is the generative type—the composite carries a different composition tree, so it is a different observable rather than a neglected term of the same one[II].

The odd permutation carrying $1 \leftrightarrow 3$ directly is the longest permutation w_0, which reverses the three generations. Its gallery crosses all three walls t_{12}, t_{23}, t_{13}, with total cost $1+2+1 = 4$—the reason the first non-zero term is of order ε^4. The coefficient 2 arises because w_0 has exactly two shortest decompositions (galleries), exchanged by point-line duality. The assignment of ε per wall and this count are proved as theorems under the specified mapping rule[II].

The parity symmetry $V(\phi) = V(-\phi)$ suppresses the direct $0 \leftrightarrow 2$ flavour transition, but it does not by itself fix the QCD vacuum angle: parity alone leaves $\theta \in \{0, \pi\}$. The observable strong-CP angle is

$$\bar{\theta} = \theta_{\mathrm{QCD}} + \arg\det M_q$$

That both terms vanish is shown by the strong-CP matching theorem of Paper II: applying A1–A3 so as to preserve the action, the fermion determinant and the realisation of large gauge transformations gives $\arg\det(M_u M_d) = 0$ and a vanishing bare θ_{QCD}, hence $\bar{\theta}(\mu_0) = 0$ at the matching point. What removes the bare θ_{QCD} is that the only unitary scalar preserving the positive measure ray is 1 (the measure-line triviality[II])—not V-parity on its own.

However, this is a statement at the matching point and not an exact zero at all scales. For the infrared $\bar{\theta}_{\mathrm{IR}}$, which includes weak CKM loops and thresholds, Paper II gives the conditional bound $|\bar{\theta}_{\mathrm{IR}}| < 6.0 \times 10^{-17}$ at the declared orders—weak $O(G_F^2)$, chiral $O(p^2)$, in the large-N_c chiral-perturbation framework; the all-orders evaluation is a continuing computation. Consequently, beyond a PQ-type axion—excluded by the absence of a PQ-type global U(1)—no claim is made about the non-existence of axion-like particles in general.

Yet CP violation *is* observed in the CKM sector ($J = 3.16 \times 10^{-5}$ [22]). This is consistent with the bare $\theta_{\mathrm{QCD}} = 0$: the CKM phase arises at higher order in ε:

$$J = \frac{\pi\varepsilon^5}{N_{\text{flags}}} = \frac{\pi\varepsilon^5}{52} = 3.06 \times 10^{-5}$$

($N_{\text{flags}} = 52$ is the total number of flags, §4.1; a theorem under the specified mapping rule[II].) The matching-point condition remains exactly zero.

The flavour and strong-CP results are summarised:

Quantity	Prediction	Experiment
V_{ub} (tree)	0; full form $2\varepsilon^4(1-2\varepsilon/3)(1-3\varepsilon/26) = 0.00384^{\text{II}}$	$\lvert V_{ub}\rvert = 0.004$ (generated at higher order)
V_{cb}	$\varepsilon^2(1-2\varepsilon/3) = 0.0410^{\text{II}}$	$\lvert V_{cb}\rvert = 0.041$
Bare θ_{QCD}	**Not a consequence of V-parity** (matching-point value $\bar{\theta}(\mu_0) = 0$; strong-CP matching theorem[II])	— (the bound $\lvert\bar{\theta}_{\text{IR}}\rvert \lesssim 10^{-10}$ applies to the infrared quantity; not directly comparable)
CKM CP violation	$J = \pi\varepsilon^5/N_{\text{flags}}$ (theorem under the mapping rule[II])	3.06×10^{-5} (expt $3.16{\times}10^{-5}$)

7.4 Neutrinos

§7.1–7.3 derived charged-fermion mass ratios and mixing angles. What remains are the neutrinos.

Mass predictions. Neutrinos are colour singlets with $\kappa = 1$, coupling to $V = -H$—the same potential as charged leptons. The mass ratio R depends only on the well shape (IPR and virial ratios), not on particle species. The same well gives the same R:

$$R_\nu = R_{\text{lepton}} = 1.529$$

This is the input-free prediction of this section. Together with the spectral ratio $(m_3/m_2)^2 = 169/5$ established below, it fixes the *dimensionless* mass ray completely, with no oscillation input at all.

Absolute masses in meV require one further ingredient, a single dimensionful scale, and exactly one is used here: the solar splitting [28] $\Delta m^2_{21} = 7.537 \times 10^{-5}$ eV^2. The atmospheric splitting is deliberately *not* used, so that the ratio $\Delta m^2_{31}/\Delta m^2_{21}$ is left on the verification side as a prediction (below). Conditional on the normal-ordering branch, setting the absolute scale with this single quantity gives[10]:

Quantity	Prediction	Status
m_1	0.312 ± 0.002 meV	Unmeasured
m_2	8.687 meV	Consistent
m_3	50.51 meV	Consistent
$\sum m_i$	59.50 ± 0.39 meV	Below Planck (< 120)

What the flag-Laplacian spectrum fixes exactly is

$$\left(\frac{m_3}{m_2}\right)^2 = \frac{169}{5}, \qquad \frac{m_3}{m_2} = \frac{|\mathrm{PG}|}{\sqrt{N+2}} = \frac{13}{\sqrt{5}} = 5.814$$

[10]The quoted uncertainties propagate from Δm^2_{21} alone; since the mass ray is fixed internally, all three masses scale as $\sqrt{\Delta m^2_{21}}$. The theory has no free parameters. Normal ordering is an input condition in this section; its independent derivation is given as a theorem[II] (a consequence of $R_\nu > 1$ and $(m_3/m_2)^2 = 169/5 > 1$). These values are consequences of setting the absolute scale, not input-free predictions, and are not counted as independent predictions.

The numerator 169 is fixed by two elementary facts that follow from the spectrum of §5.3 alone. First, the non-zero eigenvalues $4 \mp \sqrt{3}$ form a conjugate pair whose product reproduces the size of the projective plane exactly,

$$(4-\sqrt{3})(4+\sqrt{3}) = 16-3 = 13 = |\mathrm{PG}(2,\mathbb{F}_3)|, \tag{25}$$

so no separate normalisation enters. Second, write ς for the involution exchanging the eigenvectors v_1, v_2 of that pair and consider the doublet $\psi = \cos\theta_H\, v_1 + \sin\theta_H\, v_2$ (the coefficient angle θ_H is distinct from the atmospheric angle θ_{23}). Exchange invariance $\varsigma\psi = \pm\psi$ forces equal weights $|\cos\theta_H| = |\sin\theta_H|$, so the coherent amplitude has magnitude $|13\sin 2\theta_H| = 13$ regardless of the choice of sign. Reading this amplitude as the intensity $13^2 = 169$ is the Born step, justified by the observation-representation theorem of Paper II.

The denominator $5 = N+2 = 3+2$ is the rank of the solar-channel normalisation, and this count—the exhaustiveness of the carrier inventory—is the one step that uses the carrier enumeration of Paper II. The mass-squared-difference ratio, obtained by eliminating $t = m_1/m_2$ using $R_\nu = R_{\mathrm{lepton}}$, is given by the full relation$^{\mathrm{II}}$

$$\frac{\Delta m^2_{31}}{\Delta m^2_{21}} = \frac{169/5 - t^2}{1-t^2} = 33.842\ldots \quad \text{(displayed as 33.84)}$$

With non-zero m_1, 169/5 is not the exact value of the mass-squared-difference ratio; the full relation above gives 33.84, with no oscillation input. The measured ratio is $2.521/0.07537 = 33.45$, so the theoretical and measured values differ by 1.2%. Because the atmospheric splitting was not used to set the absolute scale, this difference is not absorbed anywhere: it stands as the quantity under test. Equivalently, this setting predicts $\Delta m^2_{31} = 2.551\times10^{-3}$ eV^2 against the measured 2.521×10^{-3} eV^2, and the masses in the table above satisfy the internal ratio $m_3/m_2 = 13/\sqrt{5} = 5.814$ exactly.

The calculation of this section, with the absolute scale set by Δm^2_{21}, does not itself test the mass ordering, but $\sum m_i > 70$ meV creates tension. Since $\Sigma m_i = 59.50$ meV lies close to the general normal-ordering minimum (58.89 meV at $m_1 \to 0$), a CMB-S4 [27] ($\sigma \sim 15$ meV) detection near this value would confirm the normal ordering but not distinguish this framework from the general minimal scenario; the sharp, framework-specific neutrino prediction is the ratio $\Delta m^2_{31}/\Delta m^2_{21} = 33.8$, testable by JUNO in 2026–27.

Mixing angles. In §7.3, CKM mixing angles were obtained from $\varepsilon = 0.219$—small values ($\sin\theta_C \approx 0.22$). Yet PMNS mixing angles are large ($\sin^2\theta_{12} \approx 1/3$). Why does the same $V = -H$ produce both small and large mixing?

The answer lies in colour charge. Quarks carry colour and reside in the deep well $V = -3H$, which places their mixing in the perturbative regime of small mode overlap—this is why quark mixing is small, and its expansion parameter is the generation-sector marginality ε (§3).

Neutrinos are colour-neutral; instead, the geometric allocation of gauge directions on $\mathrm{PG}(2,3)$ directly gives values corresponding to the mixing angles. The solar angle is the fraction of one complete electroweak line ($N+1 = 4$ points) in the 13 directions of PG:

$$\sin^2\theta_{12} = \frac{N+1}{|\mathrm{PG}|} = \frac{4}{13} = 0.3077 \qquad \text{(experiment: 0.3088, 0.4\%)}. \tag{26}$$

At LO the atmospheric angle is the fraction that the disjoint union of the N pure-weak directions and another line through the reference point ($N+1$ points) occupies among the 13 directions of PG, namely 7/13. With the PG-scale NLO correction ($k = |\mathrm{PG}| = 13$):

$$\sin^2\theta_{23} = \frac{7}{13}\Big(1 + \frac{\varepsilon}{13}\Big) = 0.5475 \qquad \text{(second octant)}. \tag{27}$$

The LO fraction 7/13 exceeds 1/2, predicting the second octant. In NuFIT 6.1 [28] the global best fit has moved to the first octant ($\sin^2\theta_{23} = 0.470$), but the octant is unresolved and the prediction 0.5475 lies within the 3σ range (0.432–0.587); the second-octant test is left to DUNE / Hyper-Kamiokande. The second-octant local minimum of the NuFIT 6.1 χ^2 profile lies at 0.550, and the prediction deviates from it by 0.45%.[11]

The reactor angle is $\sin^2\theta_{13} = 1/(N \cdot |\mathrm{PG}|) = 1/39$ with NLO correction ($k = N = 3$):

$$\sin^2\theta_{13} = \frac{1}{39}\Big(1 - \frac{2\varepsilon}{3}\Big) = 0.0219 \qquad (\text{expt: } 0.02249,\ 2.6\%). \tag{28}$$

The three PMNS mixing-angle predictions are summarised below:

Angle	LO	NLO prediction	Experiment	Accuracy
$\sin^2\theta_{12}$	$4/13 = 0.308$	—	0.3088	0.4%
$\sin^2\theta_{23}$	$7/13 = 0.538$	0.5475	0.550 (second-octant local min.)	0.45%
$\sin^2\theta_{13}$	$1/39 = 0.026$	0.0219	0.02249	2.6%

Sum rule: $7/13 = 3/13 + 4/13$ (at LO, $\sin^2\theta_{23} = \sin^2\theta_W + \sin^2\theta_{12}$) is a finite-counting identity with denominator 13. This is a formal sum, not a partition of point sets: the 3 pure-weak points are contained in the 4 points of the complete electroweak line (the two sets overlap; their union is 4 points). The geometric origin of the atmospheric 7 directions is the disjoint union above. NLO corrections break this sum rule; Paper II gives its form.

The PMNS and CKM phases. The atmospheric holonomy ratio $q_{23} = (7/13)(1 + \varepsilon/13) = \sin^2\theta_{23}$ also fixes the PMNS phase directly, proved as a theorem in Paper II:

$$\delta_{\mathrm{PMNS}} = 2\pi q_{23} = 197.1^\circ \quad (\text{DUNE / Hyper-Kamiokande, testable} \sim 2030) \tag{29}$$

The CKM phase does not reduce to a single closed-form expression of ε alone: it is fixed by embedding the three mixing quantities of §7.3—the screened $\sin\theta_C$, $|V_{cb}|$, and $|V_{ub}|$—together with the CP-odd invariant $J = \pi\varepsilon^5/N_{\mathrm{flags}}$ in one unitary CKM matrix and solving $\sin\delta = J/(c_{12}s_{12}c_{23}s_{23}c_{13}^2 s_{13})$ together with $\cos\delta > 0$ (the sign fixed by the initial path lifting of A1). This is proved as a theorem in Paper II and gives

$$\delta_{\mathrm{CKM}} = 62.595^\circ, \qquad \beta = 23.972^\circ, \qquad \gamma = 62.560^\circ \tag{30}$$

with the experimental values $\delta_{\mathrm{CKM}} = 66.1^\circ$ and $\gamma = 66.4^\circ$ [22].

Dirac particles. Whether neutrinos are Dirac or Majorana is an unresolved experimental question. This framework gives a clear prediction. Every fermionic operator entering the construction is a function of the number operator $\hat{n}$—the sufficient statistic of the Bernoulli family—and is therefore invariant under the phase rotation $a \to e^{i\theta}a$: the anomaly-free combination $\mathrm{U}(1)_{B-L}$ is exact, and the Majorana bilinear $aa + a^\dagger a^\dagger$ ($\Delta(B-L) = 2$) lies outside this operator algebra. V-parity then pairs the distinct states ν and ν^c (lepton number $L = \pm 1$) into a Dirac fermion (operator-level proof: the Dirac-selection theorem of Paper II [20]). Therefore neutrinoless double beta decay is exactly zero:

$$0\nu\beta\beta = 0 \qquad (\text{nEXO [29], testable} \sim 2030) \tag{31}$$

[11] The local minimum is obtained from the released grid by projecting over Δm^2_{31} and δ_{CP}; it lies at $\Delta\chi^2 \simeq 1.03$ above the global minimum, with a grid step of 0.005 in $\sin^2\theta_{23}$. Unless stated otherwise, all NuFIT 6.1 values quoted in this paper are the without-SK-atm, normal-ordering set, consistent with the Δm^2 inputs used above.

Scope of the mass formula. The mass formula of §7.1 determines mass ratios within a sector; its overall proportionality constant is sector-dependent and is not derived in this paper (in Paper II as well, this constant is treated as setting the absolute scale and does not fix the inter-sector ratios). In particular, neutrinos and charged leptons occupy the same $\kappa = 1$ well, so the framework predicts $R_\nu = R_{\text{lepton}}$ but is silent, at this stage, on the dimensionless inter-sector ratio $m(\nu_3)/m(\tau) \sim 3 \times 10^{-11}$. The origin of the neutrino Yukawa suppression is an open problem of the reconstruction, not a prediction of it.

With this, fermion mass ratios, mixing angles, and neutrino properties have been obtained from the structure of $V = -H$ and $\mathrm{PG}(2, \mathbb{F}_3)$ (the absolute masses with normal ordering assumed and the absolute scale set by Δm^2_{21}). Where the Standard Model treats these as independent free parameters, this framework treats them all as different aspects of the algebraic expansion of a single equation $V = -H$.

8 Connection to Spacetime

All calculations in §2–7 were performed in internal space (the ϕ coordinate). Neither space nor time appeared. How, then, does this zero-dimensional theory connect to four-dimensional spacetime? This chapter answers in three steps: the dimension and signature $(3, 1)$ of spacetime (§8.1), the independence of internal space and spacetime (§8.2), and why the zero-dimensional algebraic values correspond to the four-dimensional experimental values (§8.3).

8.1 Spacetime Dimension and Gravity

Section 3.4 derived $d_{\text{space}} = 3$. But "why four-dimensional spacetime?" is a separate question—it is not obvious that time is one-dimensional. We show that two independent paths give the same $D = 4$—both are consistency checks with stated premises (hereafter we distinguish the spatial dimension d from the spacetime dimension $D = d + 1$).

The first path comes from gravitational degrees of freedom. A massless spin-2 particle (graviton) in D-dimensional spacetime has $D(D-3)/2$ physical degrees of freedom (following from Poincaré-group representation theory, without assuming any particular D or Einstein's equations). Meanwhile, the $N = 3$ bound states of $V = -H$ provide $N-1 = 2$ independent internal degrees of freedom (since three probabilities sum to one). As a consistency check that takes Poincaré symmetry, a spin-2 field, and gauge reduction as premises, matching the spacetime dynamical degrees of freedom (graviton polarisations) one-to-one with the internal degrees of freedom gives:

$$D(D-3)/2 = N - 1 = 2 \quad \Longrightarrow \quad D = 4. \tag{32}$$

The second path comes from projective geometry. The projective plane $\mathrm{PG}(2, \mathbb{C}) = \mathbb{C}P^2$ is a real 4-dimensional manifold, and for $N = 3$:

$$D = \dim_{\mathbb{R}}(\mathbb{C}P^{N-1}) = 2(N-1) = 4. \tag{33}$$

The two paths are summarised:

Path	Input	Equation	Result
Graviton DOF	Spin-2 polarisations	$D(D-3)/2 = N-1 = 2$	$D = 4$
Complex carrier	Complex carrier $\mathbb{C}P^2$	$\dim_{\mathbb{R}}(\mathbb{C}P^{N-1}) = 2(N-1)$	$D = 4$

Both are consistent only for $N = 3$.

These two paths are independent—one from field-theoretic representation theory (spin-2 polarisations), the other from pure algebraic geometry (complex carrier $\mathbb{C}P^2$). That both give $D = 4$ is a non-trivial consistency check of $N = 3$.

Even with $D = 4$ fixed, the signature could be $(4, 0)$, $(3, 1)$, or $(2, 2)$. The finite field $\mathbb{F}_3$ (from the primality of $N = 3$) and the complex carrier $\mathbb{C}^3$ (from the unitary realisation) are independent constructions moored to the same $N = 3$—since the characteristics differ, no field homomorphism $\mathbb{F}_3 \to \mathbb{C}$ exists, and no lift via $\mathbb{F}_9$ succeeds either. What links the two is the correspondence between the exponent labels of the Heisenberg–Weyl system ($\mathbb{F}_3$ side) and the representation carrier ($\mathbb{C}^3$ side); what is preserved is the Weyl commutation relations and the MUB structure. $\mathbb{C}^3$ carries a complex structure from the outset. Furthermore, the transfer action $S = \int [\frac{1}{2}\dot{\phi}^2 + V]\, d\tau$ of the internal coordinate ϕ opened by A2 (§2.2) has a single evolution parameter τ, so one of the four real directions is distinguished as the evolution direction.

The complex structure and the single evolution direction alone do not uniquely fix the signature. $(3, 1)$ is fixed as the connection to the determinant form on $M_2(\mathbb{C})_{sa}$ and the structure $\mathrm{SL}_2(\mathbb{C}) \to \mathrm{SO}^+(1, 3)$.

Indeed, an element of $M_2(\mathbb{C})_{sa}$ is parametrised as

$$X = \begin{pmatrix} x^0 + x^3 & x^1 - ix^2 \\ x^1 + ix^2 & x^0 - x^3 \end{pmatrix}, \qquad \det X = (x^0)^2 - |\boldsymbol{x}|^2$$

and the determinant is precisely a quadratic form of signature $(+, -, -, -)$. The $\mathrm{SL}_2(\mathbb{C})$ action $X \mapsto AXA^\dagger$ preserves this determinant, giving $\mathrm{SO}^+(1, 3)$.

In four dimensions, Lovelock's theorem [30] uniquely determines the gravitational field equation: the only differential equation of second order or lower is the Einstein equation

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}.$$

Jacobson [31] showed that the same equation follows from $\delta Q = T_H\, dS$ on a causal horizon. Since the entropy of the present theory is $S = H(P)$, both the geometric path ($N = 3 \to D = 4$ plus Lovelock) and the thermodynamic path (Jacobson applied to $S = H$) lead from the structure of binary entropy to Einstein gravity.

8.2 Why Internal Space and Spacetime Are Independent

The coordinate $\phi = \mathrm{logit}(\sigma)$ lives in internal space; the coordinate x lives in spacetime. Why are they independent?

The answer lies in the construction of $V = -H$. The potential $V = -H(\sigma(\phi))$ is built as the internal zero-mode of the Bernoulli degree of freedom and contains no explicit spacetime coordinate x. However, the coordinate-independence $\partial V/\partial x = 0$ alone does not exclude a mixed kinetic term. What makes the decomposition hold is the construction in which ϕ (internal) and x (spacetime) are independent degrees of freedom, with generators acting on separate factors and no cross term. Under this, the full Hamiltonian takes the form $\hat{H} = \hat{H}_\phi + \hat{H}_x$, and the Hilbert space decomposes as a tensor product $\mathcal{H}_{\mathrm{int}}(\phi) \otimes \mathcal{H}_{\mathrm{space}}(x)$.

$$\hat{H} = \hat{H}_\phi \otimes \mathbf{1} + \mathbf{1} \otimes \hat{H}_x, \qquad \mathcal{H} = \mathcal{H}_{\mathrm{int}}(\phi) \otimes \mathcal{H}_{\mathrm{space}}(x)$$

Since this composite space is separable with dim ≥ 3 ($\mathcal{H}_{\mathrm{int}}(\phi)$ is 3-dimensional from the generation factor alone, $\mathcal{H}_{\mathrm{space}}(x)$ is infinite-dimensional), Gleason's theorem [32] restricts every probability assignment additive over orthogonal projections to the trace form $p(P) = \mathrm{Tr}(\rho P)$—the form of the Born rule is not an axiom but a consequence of the tensor-product structure. What is unique is the form of the assignment; the single detection probability $\sigma = \langle n \rangle$ of §2 does not by itself determine the state ρ.

Two observational facts confirm this. First, all gauge couplings are independent of generation—direct evidence that ϕ and x are not coupled. Second, the 0D theory computes only dimensionless quantities (mass ratios, mixing angles), while dimensionful quantities (absolute masses) require spacetime—precisely the prediction of the tensor-product structure.

The analogy with spin illuminates this structure. Spin algebra is finite-dimensional and independent of the Lagrangian, yet determines 4D physical quantities (magnetic moment, fine structure). Generations have the same structure—the finite-dimensional $V = -H$ corresponds to 4D mass ratios and mixing angles.

Once both factors are in place, the form of the 4D field theory is determined as well. The spacetime factor carries the local Lorentz carrier of §8.1 (spinor fields); the internal factor carries the finite data of §4—the gauge algebra with its representations, the single Higgs doublet, and the four Yukawa edges. The first-order local operator coupling the two is narrowed to a single form compatible with the tensor-product structure and the internal data:

$$\mathcal{D}_{A,\Phi} = i\gamma^\mu(\nabla_\mu + A_\mu) + \gamma_5 \otimes \Phi \tag{34}$$

where A_μ is the gauge connection and Φ contains only the single Higgs and the four Yukawa edges (the classification of §4).

The interactions then decompose uniquely as the curvature components of this single operator—the field strength F_A (gauge self-interaction), the covariant derivative $D_A\Phi$ (the gauge–Higgs coupling), and $\Phi^2 - \Phi_0^2$ (the Higgs potential)—whose magnitude gives the bosonic action, while $\langle \Psi, \mathcal{D}_{A,\Phi}\Psi \rangle$ gives the fermionic action. Each term of the Standard-Model Lagrangian is thus not a separately postulated assumption but a curvature component of a single first-order coupling.

The rigorous uniqueness of this form (the exclusion of other coupling types) and the values of the dimensionless coefficients are given, under the stated realisation conditions, by the manifestation theorem of Paper II. The transport of the electromagnetic clock is unique, as a continuous map preserving semigroup composition, up to a single scale-setting quantity[II]. The β functions of individual renormalisation schemes and the numerical transport between schemes are the continuing computation that displays the same physical quantity in the coordinates of each scheme, and the absolute scale v is what carries the derived quantities into physical units—the conventional layer (both in Paper II).

8.3 Correspondence Between 0D Values and 4D Experiment

The tensor-product structure explains why 0D algebraic values coincide with tree-level values of the 4D renormalised theory. The 0D values are algebraic constants. In 4D, tree level is the limit of zero loop corrections—i.e. the limit in which spacetime dynamics does not affect internal structure—which is precisely the independence guaranteed by $\mathcal{H}_{\mathrm{int}}(\phi)\otimes\mathcal{H}_{\mathrm{space}}(x)$. Radiative corrections arise from loop momenta in the spacetime factor $\mathcal{H}_{\mathrm{space}}$ and enter as sub-percent corrections to the internal algebraic values (§6: $\sin^2\theta_W = 3/13 + \Delta_{\mathrm{rad}} = 0.23122$). The 0D values are neither RG fixed points nor UV boundary conditions, but the unique algebraically determined point serving as the reference for comparison with 4D theory—running quantities acquire the 4D RG evolution and loop corrections, while quantities such as discrete state counts do not run.

Concretely, each observable carries a stated definition point for the comparison (α at the Thomson point; $\sin^2\theta_W$ and α_s at M_Z; §5–§6), and the 0D value is compared with the renormalised 4D value at that point. Changing the scale is described on the 4D side as RG evolution; on the internal side it corresponds to reading the same finite resolvent that gave Z in §5.3 at a different evaluation point, and the closed form of this response kernel is derived in Paper II[II]. The closed form of the β function in any renormalisation scheme is obtained by the transport computation between definition points.

For the Thomson-normalised physical electromagnetic effective charge, by contrast, the identification of the electromagnetic response together with the uniqueness of the transport yields, as a theorem, uniform boundedness on the whole Euclidean-momentum half-line and the absence of a Landau pole[II]—α can be followed along it from its Thomson-point value to a finite ultraviolet value. Poles appearing in perturbative displays are coordinate singularities created by a singular reparametrisation of the coupling, not poles of the physical current response.

With this, the relationship between internal structure (ϕ) and spacetime (x) is established. $V = -H$ gives internal values corresponding to mass ratios, mixing angles, and coupling constants. The spacetime factor supplies, in addition to the overall energy scale, the 4D RG evolution and loop corrections for running quantities.

9 Results

This chapter collects the results. Twenty parameters are derived in this paper. A further six ($|V_{cb}|$, $|V_{ub}|$, δ_{CKM}, δ_{PMNS}, m_3/m_2, $\Delta m^2_{31}/\Delta m^2_{21}$) require the specified operator words and the spectral structure of the flag Laplacian, and are derived in Paper II—their values are displayed in the tables of this paper, with the attribution marked in the § column.

Of the 26 Standard-Model parameters, the single dimensionful one — the electroweak scale v — is the external unit of the dimensionless internal theory; its specification is not a residue of the derivation but the setting of the absolute scale, the conventional layer (§8.2, Paper II). All predictions of this paper are dimensionless quantities measured in units of v (plus the absolute masses of §7.4, with the absolute scale set by Δm^2_{21}).

The full list of results is given below (experimental values from PDG 2026 [22], NuFIT 6.1 [28], and CODATA 2022 [34]; the comparison value for $1/\alpha_{\mathrm{em}}$ alone is the Cs-side CODATA 2018 value, per the Cs/Rb dispute—§5.3, Table 4). The largest deviation among the quantities derived in this paper ($\sin^2\theta_{13}$ at 2.6%) lies within about 1σ of the corresponding experimental uncertainty (§7.4). The $\sin^2\theta_{23}$ comparison is against the second-octant local minimum 0.550 of the NuFIT 6.1 χ^2 profile; the NuFIT 6.1 global best fit lies in the first octant (0.470), the octant is unresolved, and the prediction lies within the 3σ range (§7.4). The 5.3% for δ_{CKM} is a genuine residual of this sector.

Table 1. The derived equations and their values.[12]

Quantity	Equation	Value
$1/\alpha$ (leading)	$1/\alpha + \alpha = \lambda_1 + Z$ ($\lambda_1 = 4 - \sqrt{3}$, $Z = 134.77613$)	137.0368
$1/\alpha$ (all orders)	physical root of the all-orders Dyson series (§5.3)	137.035999084
$\sin^2\theta_W$	$3/13$ (next correction $+\alpha \cdot 135/2197$)	$0.23077 \to 0.23122$
α_s	$(3/26)(1+\varepsilon/10)$	0.1179
m_W/m_Z	$\sqrt{10/13}$	0.8771
m_H/v	$\frac{1}{2}\sqrt{1+2\varepsilon/13}$	0.5084
mass ratios R	$m_n \propto \lvert E_n\rvert\, \mathrm{IPR}_n^b(\lvert\langle V\rangle_n\rvert/T_n)^{\varepsilon/N}$ ($\kappa = 1, 3$)	1.529, 1.777, 2.273
down-type correction	$m_{\text{eff}} = 1 - N_c\,\sigma(1-\sigma)$	(enters R_{down})
$\sin\theta_C$	$\varepsilon(1+\varepsilon/9)$, after screening $\times(1 - 9\alpha/(26\pi))$	$0.2245 \to 0.22435$
$\lvert V_{cb}\rvert$	$\varepsilon^2(1-2\varepsilon/3)$	0.0410
$\lvert V_{ub}\rvert$	$2\varepsilon^4(1-2\varepsilon/3)(1-3\varepsilon/26)$	0.00384
J	$\pi\varepsilon^5/52$	3.06×10^{-5}
δ_{CKM}	$\sin\delta = J/(c_{12}s_{12}c_{23}s_{23}c_{13}^2 s_{13})$, $\cos\delta > 0$	62.6°
$\sin^2\theta_{12}$	$4/13$	0.3077
$\sin^2\theta_{23}$	$(7/13)(1+\varepsilon/13)$	0.5475
$\sin^2\theta_{13}$	$(1/39)(1-2\varepsilon/3)$	0.0219
δ_{PMNS}	$2\pi \cdot (7/13)(1+\varepsilon/13)$	197.1°
m_3/m_2	$13/\sqrt{5}$	5.814
$\Delta m_{31}^2/\Delta m_{21}^2$	$(169/5 - t^2)/(1-t^2)$ ($t = m_1/m_2$)	33.84
$0\nu\beta\beta$	exactly zero	0
$\bar\theta(\mu_0)$	zero at the matching point	0

Table 2. Comparison with Standard Model constants.

Parameter	Prediction	Experiment	Accuracy	§
Gauge couplings				
$1/\alpha_{\text{em}}$ ($\to g_1$) (leading; Thomson point)	137.0368	137.036	0.0006%	5.3
$1/\alpha_{\text{em}}$ (all orders)	137.035999084	137.035999084(21)	all digits (Cs)	Paper II
α_s ($\to g_3$) ($\overline{\text{MS}}$, M_Z)	0.1179	0.1180	0.07%	5.2
$\sin^2\theta_W$ ($\to g_1/g_2$) ($\overline{\text{MS}}$, M_Z)	0.23122	0.23122	$< 0.01\%$	6.3
m_W/m_Z (on-shell identification)	0.88137	0.8813	$+1.0\sigma$	Paper II
Higgs				
m_H/v ($\to \lambda$)	0.5084	0.5082	0.03%	6
Fermion masses				
R_{lepton} ($\to y_e, y_\mu, y_\tau$)	1.5293	1.5294	0.006%	7.1
Q (Koide)	1.499985	1.500005(11)	2σ	7.1
R_{up} ($\to y_u, y_c, y_t$)	1.777	1.770	0.4%	7.2
R_{down} ($\to y_d, y_s, y_b$)	2.273	2.276	0.1%	7.2
CKM / PMNS mixing				
$\sin\theta_C$ ($\to$ CKM θ_{12})	0.2245	0.2243	0.10%	7.3
$\lvert V_{cb}\rvert$	0.0410	0.0407(13)	0.8%	Paper II
$\lvert V_{ub}\rvert$	0.00384	0.00389(16)	1.2%	Paper II
δ_{CKM}	62.6°	66.1°	5.3%	Paper II
$\sin^2\theta_{12}$	4/13	0.3088	0.4%	7.4
$\sin^2\theta_{23}$	0.5475	0.550 (local min.)	0.45%	7.4
$\sin^2\theta_{13}$	0.0219	0.02249	2.6%	7.4

[12] All theoretical values in Table 1 are computed solely from the spectral and incidence data derived from $V = -H$ and $\mathrm{PG}(2, \mathbb{F}_3)$; no experimental value enters their calculation. The correspondence between the resulting mathematical structures and physical structures, and the proof of its uniqueness under the stated realisation conditions, are given in Paper II.

Table 3. Structural results (derivations in the sections; D and the signature in §8 are consistency checks under stated premises).

Structure	Result	§
Number of generations N	3 (a fourth sequential chiral generation is excluded)	3
Spatial dimension d	3	3.4
Spacetime dimension D and signature	4, $(3,1)$	3.4, 8
Gauge group	SU(3)×SU(2)×U(1)	4, Paper II
Neutrino properties	fixed by $\mathrm{U}(1)_{B-L}$ and V-parity	7.4
$\bar{\theta}(\mu_0)$	0	7.3, Paper II

Table 4. Testable predictions.[13]

Prediction	Value	Verification experiment	§
$\Delta m^2_{31}/\Delta m^2_{21}$	full relation (leading $m_3/m_2 = 13/\sqrt{5}$, non-zero m_1 included) $=$ 33.84	JUNO [26] (2026–27)	Paper II
$1/\alpha$ (all orders)	137.035999084—the Cs/Rb recoil discrepancy resolves on the Cs side	next-generation photon-recoil / $g-2$ comparisons	Paper II
θ_{23} octant	Second ($> 45°$)	DUNE / HK (∼2030)	7.4
δ_{PMNS}	$2\pi(7/13)(1+\varepsilon/13) = 197.1°$	DUNE / HK (∼2030)	Paper II
R_ν (mass ratio; input-free)	1.529 ($= R_{\mathrm{lepton}}$)	osc. + absolute mass	7.4
Σm_ν	59.50 meV (absolute scale set by Δm^2_{21}; $\approx$ NO min. 58.9)	CMB-S4 [27] (∼2030)	7.4
m_1	0.312 meV (absolute scale set by Δm^2_{21})	CMB-S4 (via Σm_ν)	7.4
$0\nu\beta\beta$	Exactly zero	nEXO [29] (∼2030)	7.4
Koide deviation δ	-1.5×10^{-5}	Belle II [33] (2027–30)	7.1

The Standard Model treats these physical quantities as independent parameters to be measured and input, but the derivation in this paper has no input parameters. All numerical results above (including every value in Tables 1–4 and the interval certifications of §3) can be independently reproduced with the verification-code suite provided as supplementary material.

10 Conclusion

This paper has shown that structures and numerical values normally treated as independent inputs in the Standard Model are reproduced, with no adjustable parameters, by a single equation fixed by the three axioms of existence. This agreement indicates that the identifications between mathematical and physical structure made throughout this paper may be correct. If so, this suggests that the fundamental laws of matter and the Universe are different aspects of the single equation

$$V = -H(\sigma(\phi))$$

[13] One further prediction on an open experimental dispute, of the same kind as the $1/\alpha$ entry: first-row CKM unitarity is exact (Paper II), and the $|V_{ud}|$ dispute—global fit 0.97431(16) versus the direct superallowed $0^+ \to 0^+$ value 0.97367(32)—resolves on the unitarity side.

and can be described as a necessity from this equation derived from the axioms of existence.

Paper II [20] treats the derivation of the remaining six parameters, the spectral proof of the gauge decomposition 9+3+1, and a more detailed account of the uniqueness of the construction used in this paper. Subsequent papers extend this framework to gravity, the Universe, and the unification of forces.

To be, or not to be — that is the equation.

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